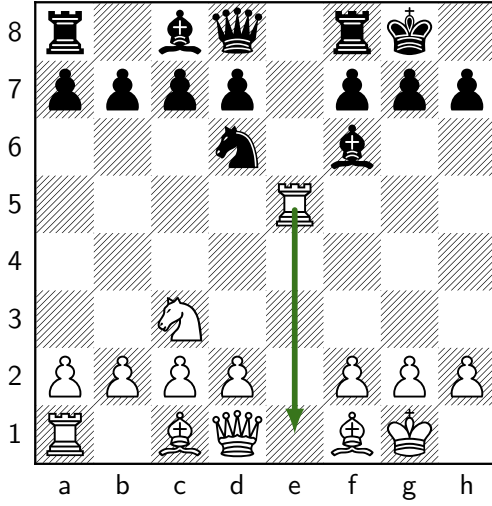


Melanie-MASTERS (corpus: human masters) — match game 5

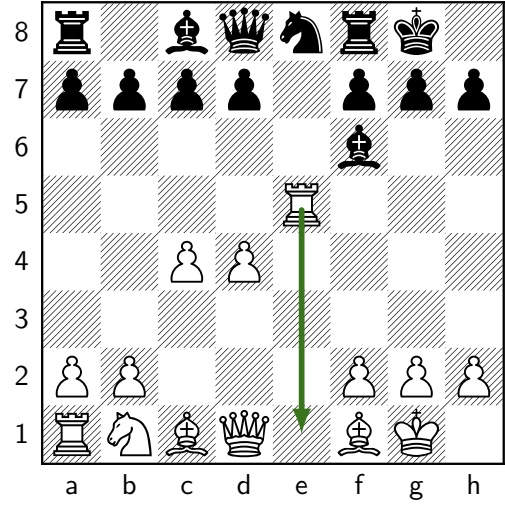
Melanie (Deep_Vision)

Abstract

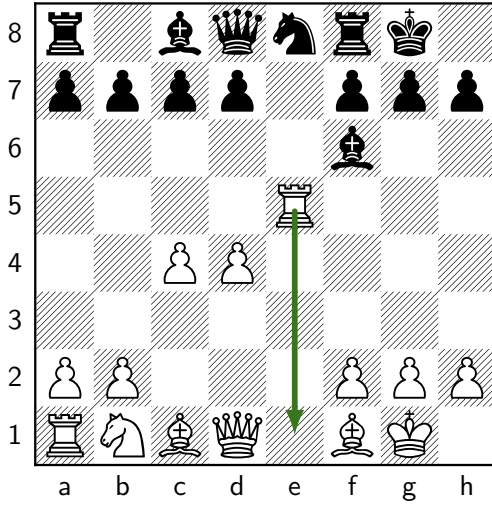
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



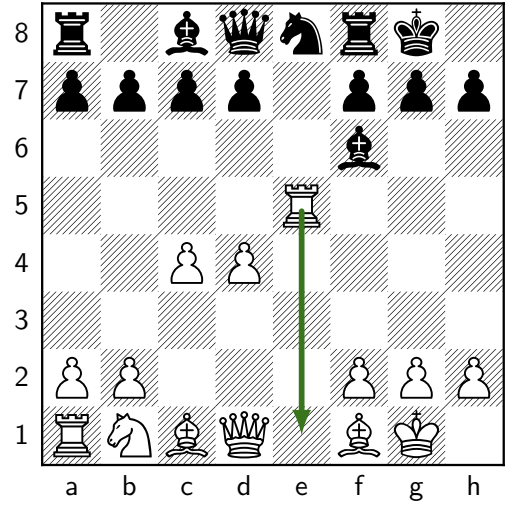
(a)



(b)

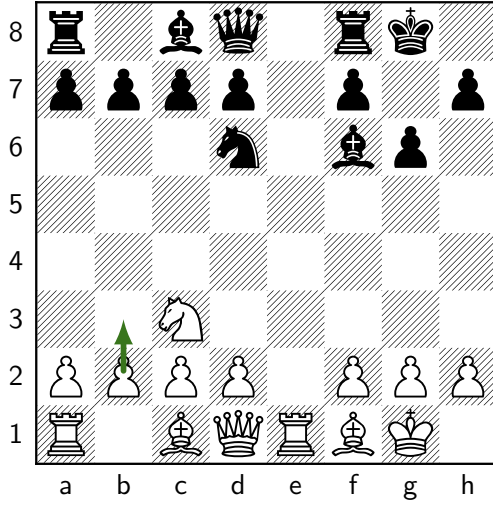


(c)

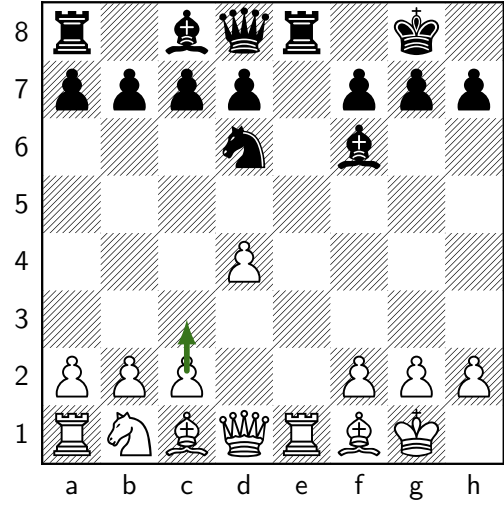


(d)

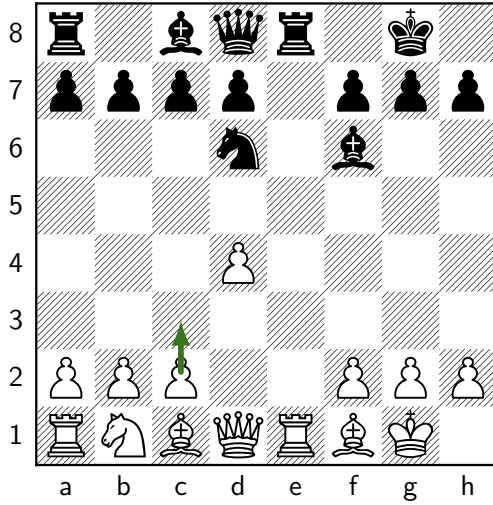
Figure 1: **Move 10, White: Re1.** (a) Our position; Melanie plays **Re1** (**e5e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#521` (ply 20) [structural distance $d=42$, QAP obj=28]: the master played **Re1**; under μ our *Re5* plays the role of that game’s *Re5*; cross-type roles: our *pc7* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move **e5e1** (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#423` (ply 20) [structural distance $d=42$, QAP obj=28]: the master played **Re1**; under μ our *Re5* plays the role of that game’s *Re5*; cross-type roles: our *pc7* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move **e5e1** (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#1414` (ply 20) [structural distance $d=42$, QAP obj=28]: the master played **Re1**; under μ our *Re5* plays the role of that game’s *Re5*; cross-type roles: our *pc7* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move **e5e1** (**endorses this move**). Synthesis: 11 retrieved analogs transferred to **Re1** (summed QAP objective 293, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



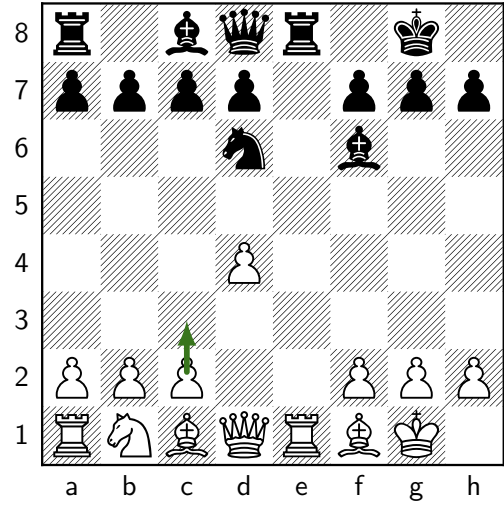
(a)



(b)

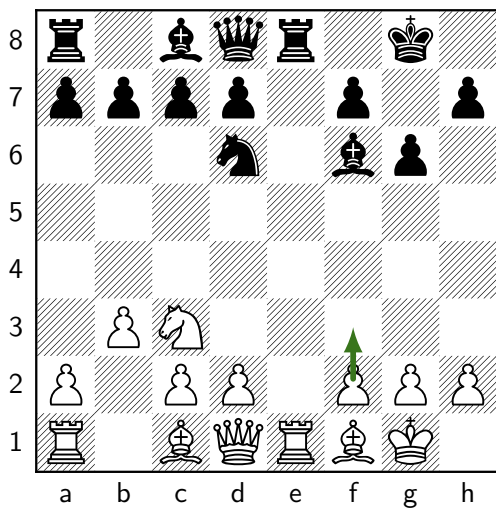


(c)

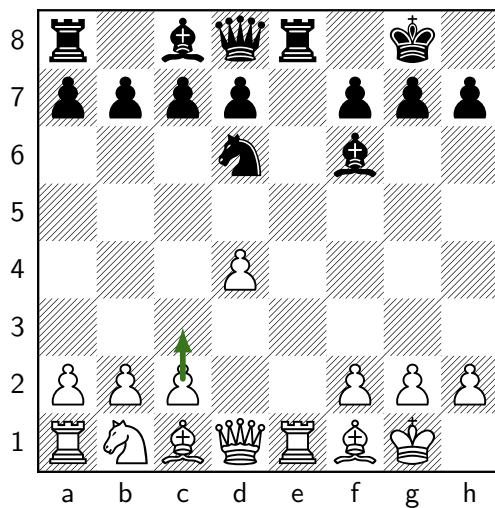


(d)

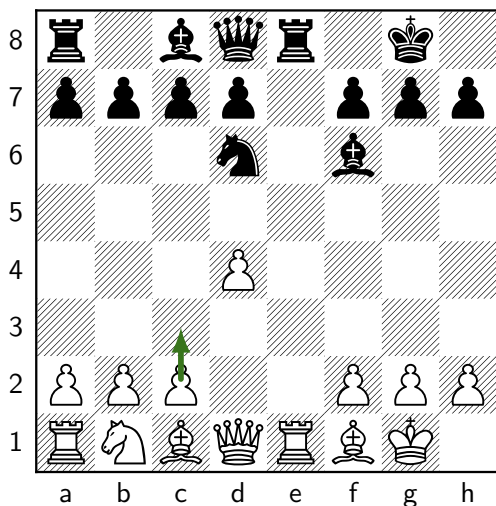
Figure 2: **Move 11, White: b3**. (a) Our position; Melanie plays b3 (b2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#389` (ply 20) [structural distance $d=55$, QAP obj=19]: the master played c3; under μ our *Pb2* plays the role of that game’s *Pc2*; cross-type roles: our *Bf1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move b2b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#512` (ply 20) [structural distance $d=55$, QAP obj=19]: the master played c3; under μ our *Pb2* plays the role of that game’s *Pc2*; cross-type roles: our *Bf1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move b2b3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#1555` (ply 20) [structural distance $d=55$, QAP obj=19]: the master played c3; under μ our *Pb2* plays the role of that game’s *Pc2*; cross-type roles: our *Bf1* as their *Ra1*, our *Ra1* as their *Nb1*; transferred move b2b3 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to b3 (summed QAP objective 259, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



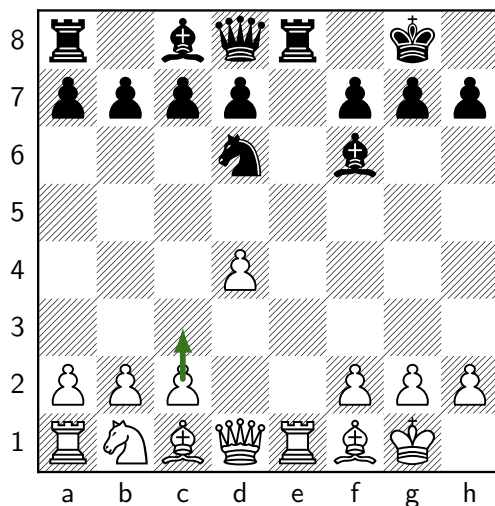
(a)



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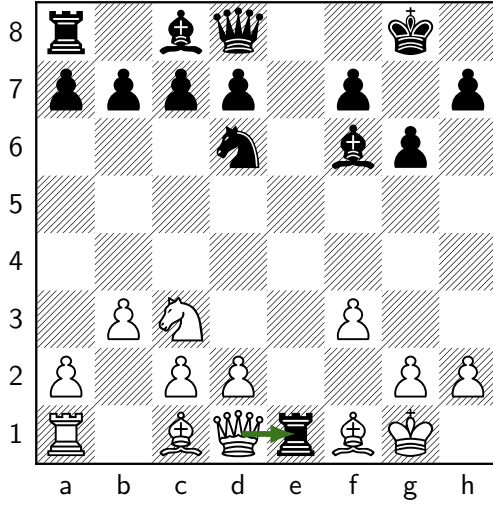


(c)

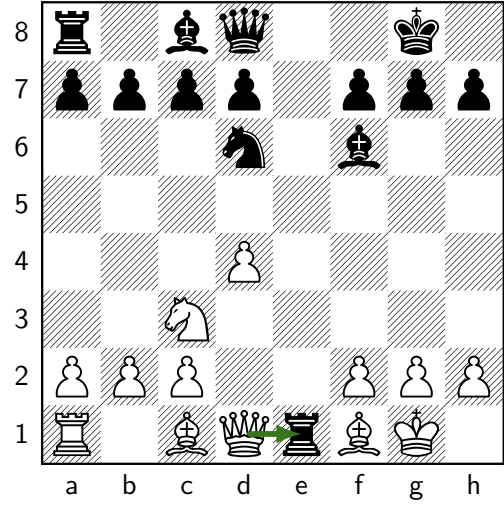


(d)

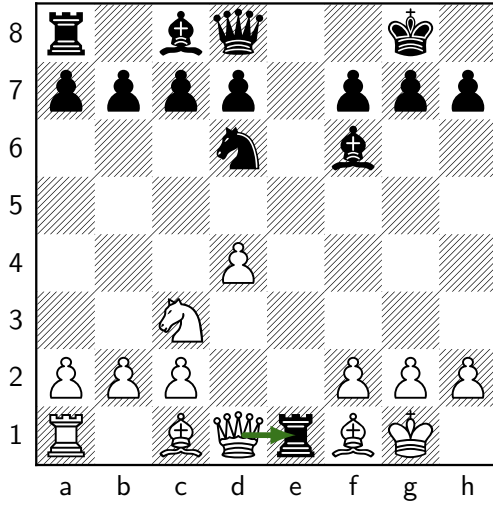
Figure 3: **Move 12, White: f3**. (a) Our position; Melanie plays f3 (f2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#389` (ply 20) [structural distance $d=55$, QAP obj=58]: the master played c3; under μ our *Pf2* plays the role of that game’s *Pc2*; cross-type roles: our *Ra1* as their *Nb1*, our *pg6* as their *Bc1*; transferred move f2f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#512` (ply 20) [structural distance $d=55$, QAP obj=58]: the master played c3; under μ our *Pf2* plays the role of that game’s *Pc2*; cross-type roles: our *Ra1* as their *Nb1*, our *pg6* as their *Bc1*; transferred move f2f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#1555` (ply 20) [structural distance $d=55$, QAP obj=58]: the master played c3; under μ our *Pf2* plays the role of that game’s *Pc2*; cross-type roles: our *Ra1* as their *Nb1*, our *pg6* as their *Bc1*; transferred move f2f3 (**endorses this move**). Synthesis: 33 retrieved analogs transferred to f3 (summed QAP objective 1909, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



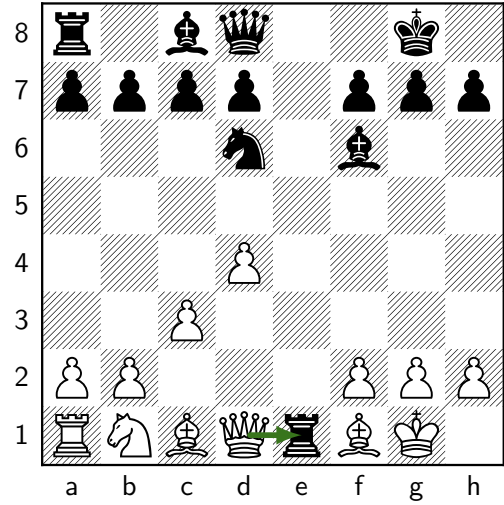
(a)



(b)

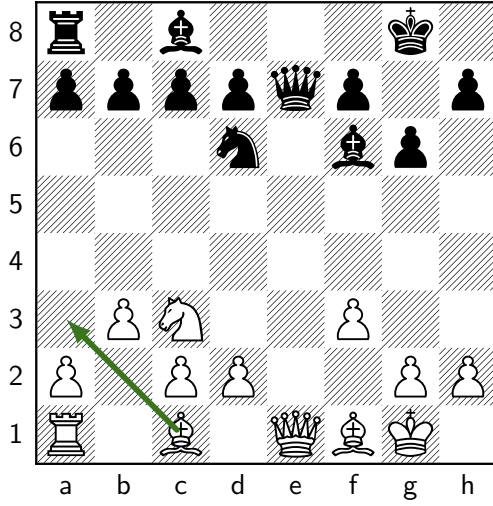


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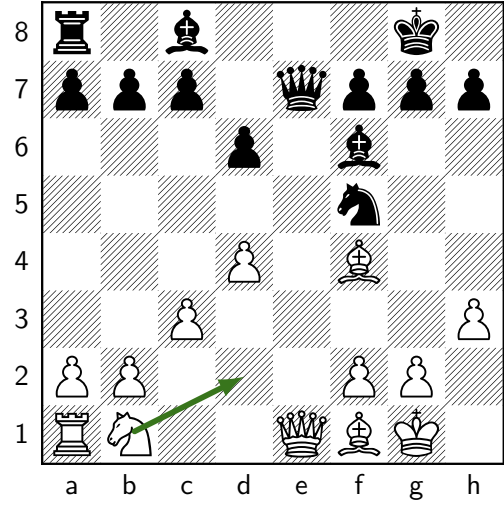


(d)

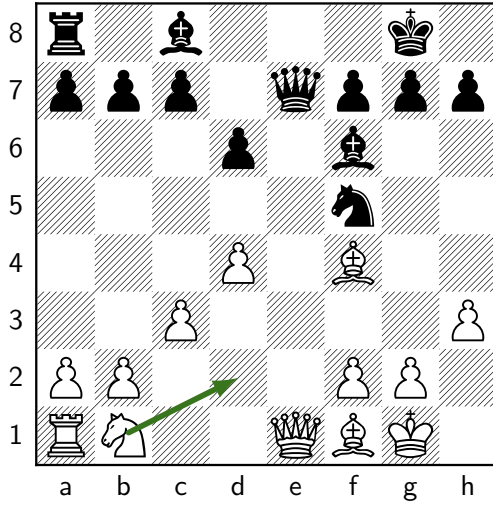
Figure 4: **Move 13, White: Qxe1.** (a) Our position; Melanie plays Qxe1 (d1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2436` (ply 22) [structural distance $d=44$, QAP obj=110]: the master played Qxe1; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pd2 as their Ra1, our bf6 as their Pb2; transferred move d1e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1605` (ply 22) [structural distance $d=44$, QAP obj=110]: the master played Qxe1; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our Pd2 as their Ra1, our bf6 as their Pb2; transferred move d1e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#389` (ply 22) [structural distance $d=47$, QAP obj=102]: the master played Qxe1; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our bf6 as their Ra1, our Pd2 as their Bc1; transferred move d1e1 (**endorses this move**). Synthesis: 40 retrieved analogs transferred to Qxe1 (summed QAP objective 4102, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



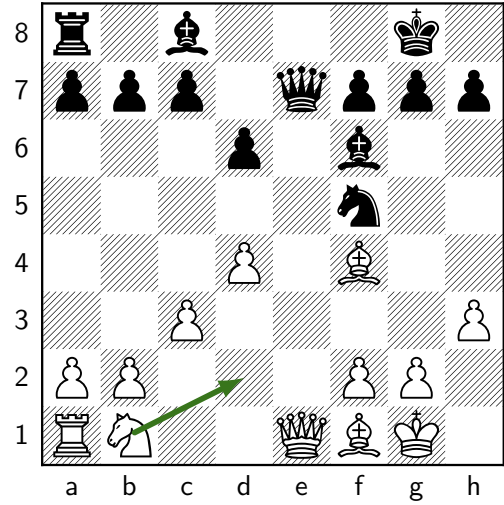
(a)



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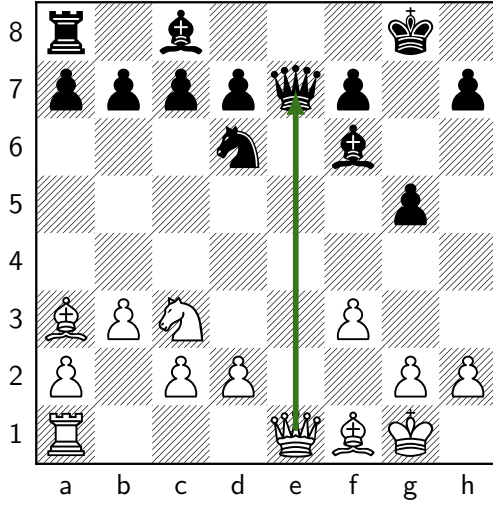


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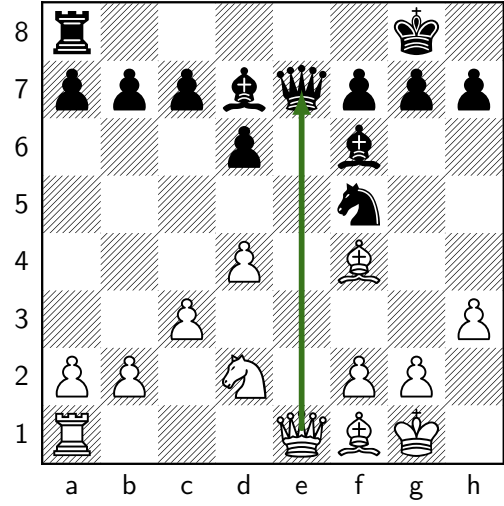


(d)

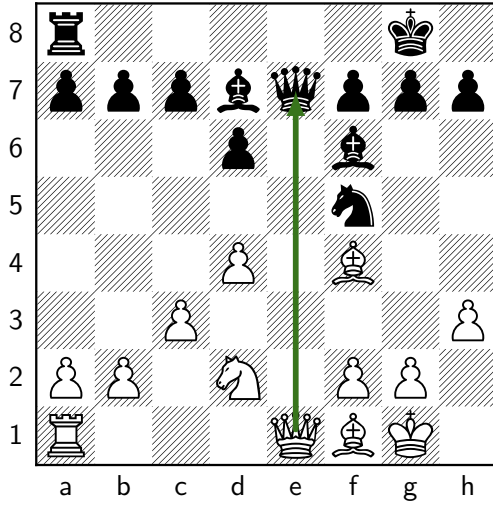
Figure 5: **Move 14, White: Ba3**. (a) Our position; Melanie plays Ba3 (c1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#2524` (ply 28) [structural distance $d=56$, QAP obj=71]: the master played Nd2; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nc3* as their *Pd4*, our *pg6* as their *Bf4*; transferred move c1a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#2690` (ply 28) [structural distance $d=56$, QAP obj=71]: the master played Nd2; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nc3* as their *Pd4*, our *pg6* as their *Bf4*; transferred move c1a3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#808` (ply 28) [structural distance $d=56$, QAP obj=71]: the master played Nd2; under μ our *Bc1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nc3* as their *Pd4*, our *pg6* as their *Bf4*; transferred move c1a3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ba3 (summed QAP objective 288, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



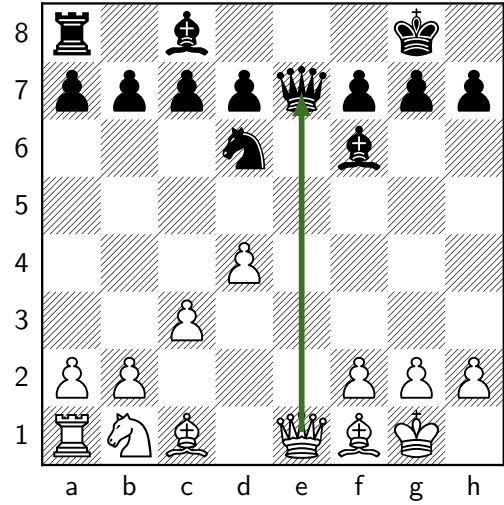
(a)



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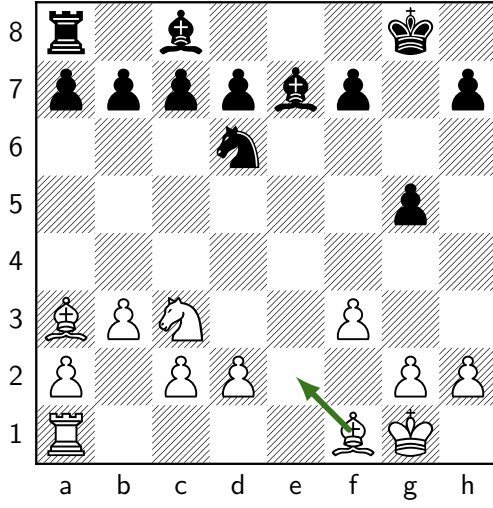


(c)

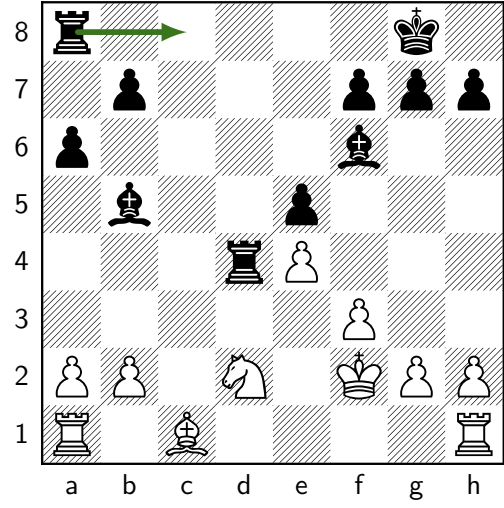


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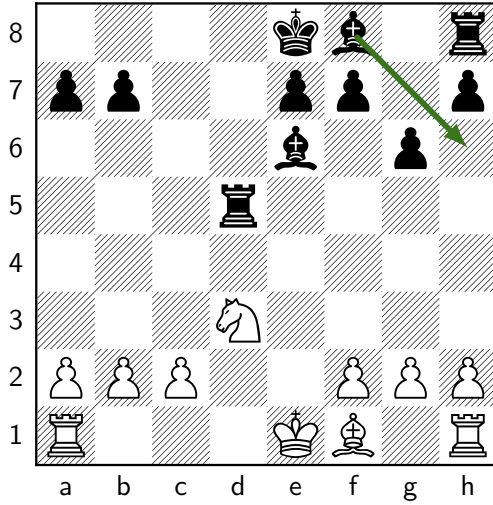
Figure 6: **Move 15, White: Qxe7.** (a) Our position; Melanie plays Qxe7 (e1e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0013#2690 (ply 30) [structural distance $d=46$, QAP obj=76]: the master played Qxe7; under μ our Qe1 plays the role of that game’s Qe1; cross-type roles: our Pf3 as their Nd2, our Nc3 as their Pd4; transferred move e1e7 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0026#1950 (ply 30) [structural distance $d=46$, QAP obj=76]: the master played Qxe7; under μ our Qe1 plays the role of that game’s Qe1; cross-type roles: our Pf3 as their Nd2, our Nc3 as their Pd4; transferred move e1e7 (**endorses this move**). (d) Retrieved from twic1500#337 (ply 24) [structural distance $d=56$, QAP obj=70]: the master played Qxe7; under μ our Qe1 plays the role of that game’s Qe1; cross-type roles: our Pa2 as their Ra1, our Ba3 as their Nb1; transferred move e1e7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qxe7 (summed QAP objective 291, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



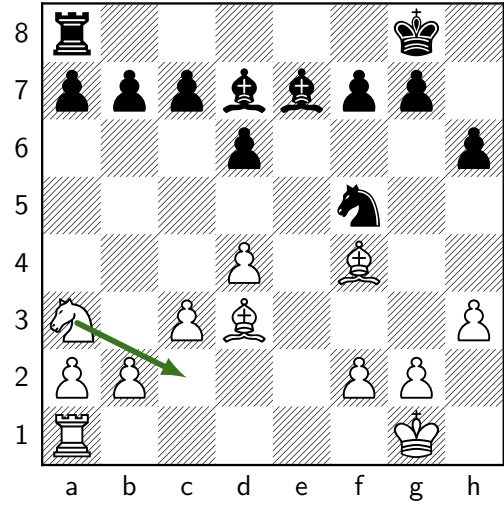
(a)



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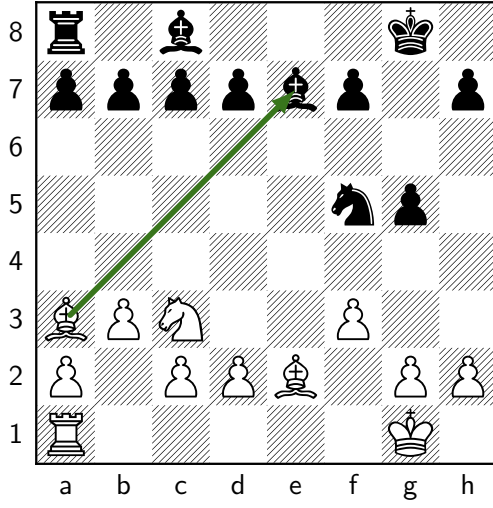


(c)

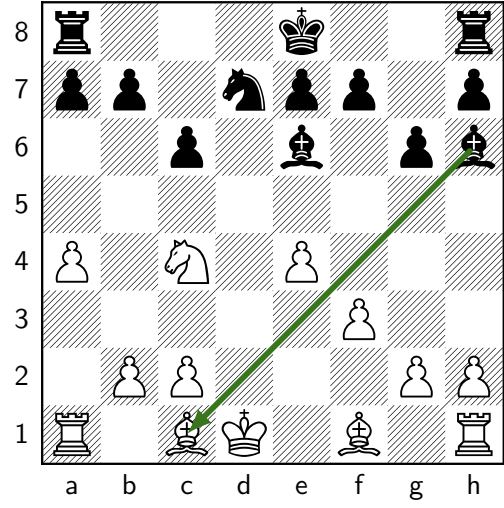


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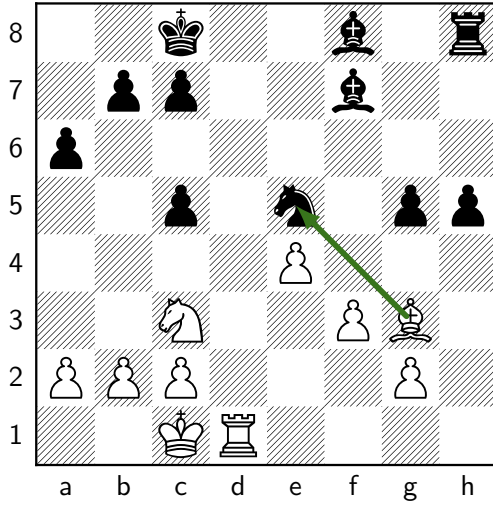
Figure 7: **Move 16, White: Be2.** (a) Our position; Melanie plays Be2 (f1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0003#71 (ply 37) [structural distance $d=70$, QAP obj=26]: the master played Rc8; under μ our Bf1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Ph7, our Pf3 as their Bf6; transferred move f1e2 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0010#3587 (ply 31) [structural distance $d=76$, QAP obj=18]: the master played Bh6; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our Pa2 as their Rh8, our ra8 as their Pf7; transferred move f1e2 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0022#214 (ply 34) [structural distance $d=68$, QAP obj=17]: the master played Nc2; under μ our Bf1 plays the role of that game’s Na3 (a **cross-type** correspondence); cross-type roles: our Pc2 as their Ra1, our ra8 as their Pc3; transferred move f1e2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Be2 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



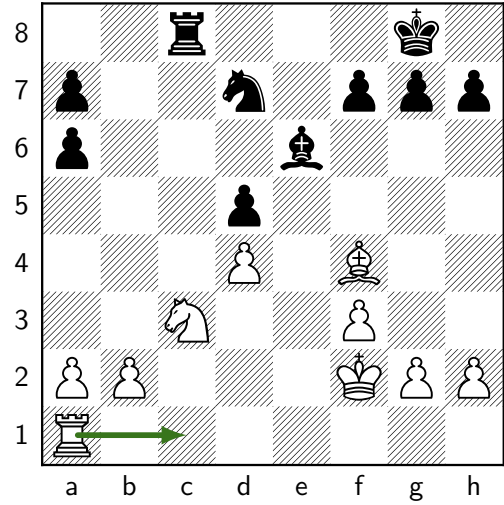
(a)



(b)



(c)



(d)

Figure 8: **Move 17, White: Bxe7.** (a) Our position; Melanie plays Bxe7 (a3e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3346` (ply 21) [structural distance $d=71$, QAP obj=19]: the master played Bxc1; under μ our Ba3 plays the role of that game’s Bh3; cross-type roles: our Pg2 as their Nd7, our pf7 as their Be6; transferred move a3e7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#1283` (ply 42) [structural distance $d=60$, QAP obj=17]: the master played Bxe5; under μ our Ba3 plays the role of that game’s Bg3; cross-type roles: our Be2 as their Rd1, our Ba3 as their Pf3; transferred move a3e7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#387` (ply 38) [structural distance $d=70$, QAP obj=28]: the master played Rc1; under μ our Ra1 plays the role of that game’s Ra1; cross-type roles: our nf5 as their Pb2, our be7 as their Nc3; transferred move a1c1. Synthesis: 2 retrieved analogs transferred to Bxe7 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

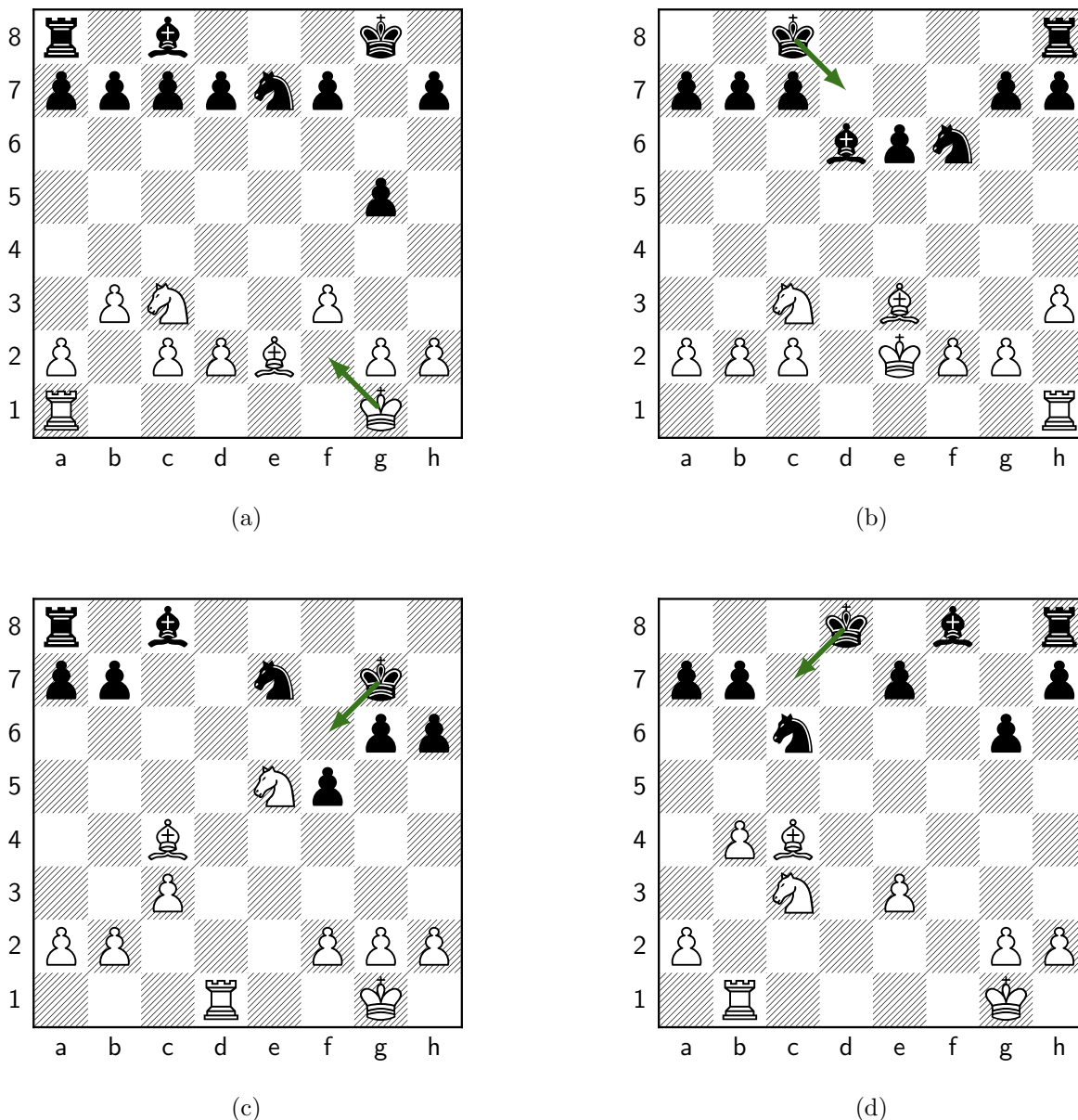
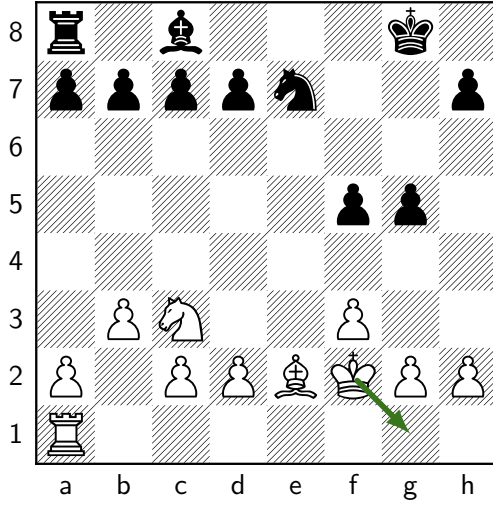
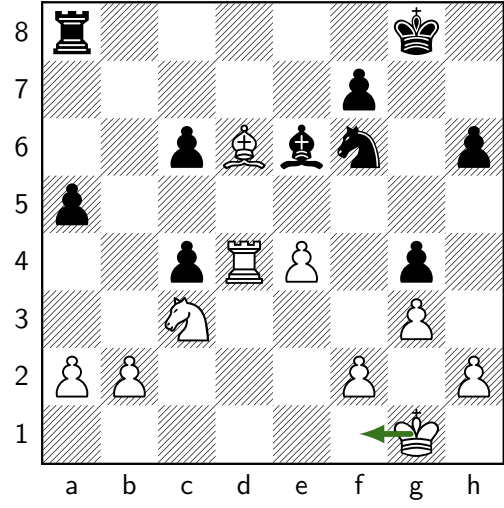


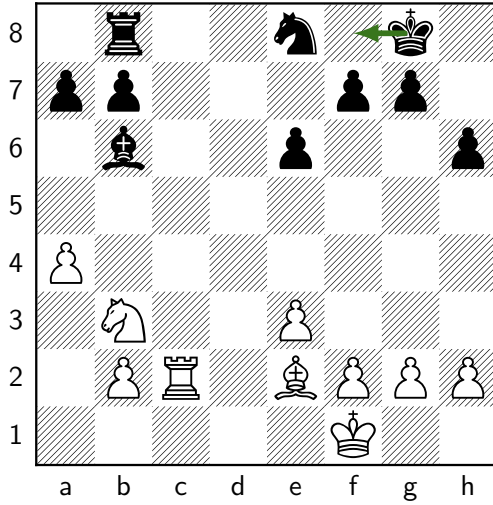
Figure 9: **Move 18, White: Kf2.** (a) Our position; Melanie plays Kf2 (g1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#2775` (ply 31) [structural distance $d=61$, QAP obj=5]: the master played Kd7; under μ our Kg1 plays the role of that game’s Kc1; cross-type roles: our Nc3 as their Pe6, our pb7 as their Nf6; transferred move g1f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#456` (ply 45) [structural distance $d=62$, QAP obj=4]: the master played Kf6; under μ our Kg1 plays the role of that game’s Kg2; cross-type roles: our Nc3 as their Ra8, our Pf3 as their Bc8; transferred move g1f2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#3150` (ply 39) [structural distance $d=58$, QAP obj=4]: the master played Kc7; under μ our Kg1 plays the role of that game’s Kd1; cross-type roles: our pg5 as their Bf8, our Be2 as their Pg6; transferred move g1f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf2 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



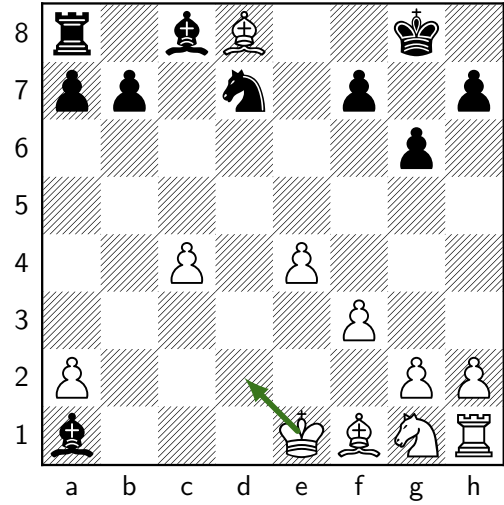
(a)



(b)

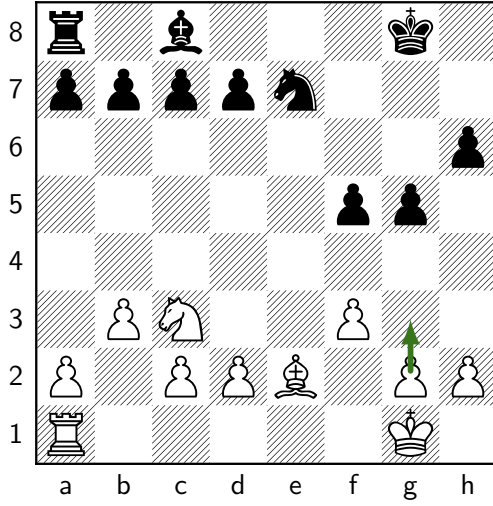


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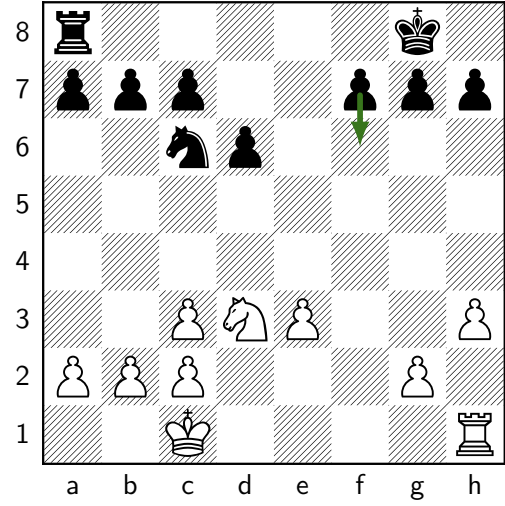


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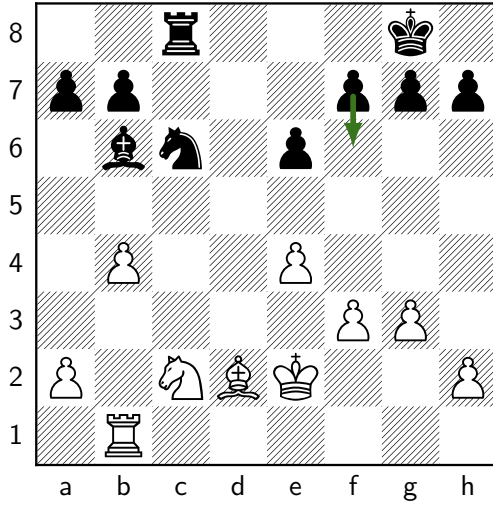
Figure 10: **Move 19, White: Kg1.** (a) Our position; Melanie plays Kg1 (f2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#1212` (ply 44) [structural distance $d=71$, QAP obj=4]: the master played Kf1; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our bc8 as their Ph2, our Pf3 as their Nc3; transferred move f2g1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2166` (ply 47) [structural distance $d=66$, QAP obj=3]: the master played Kf8; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our pa7 as their Rb8, our pb7 as their Ne8; transferred move f2g1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#1332` (ply 26) [structural distance $d=72$, QAP obj=2]: the master played Kd2; under μ our Kf2 plays the role of that game’s Ke1; cross-type roles: our Pf3 as their ba1, our pa7 as their Ng1; transferred move f2g1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg1 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



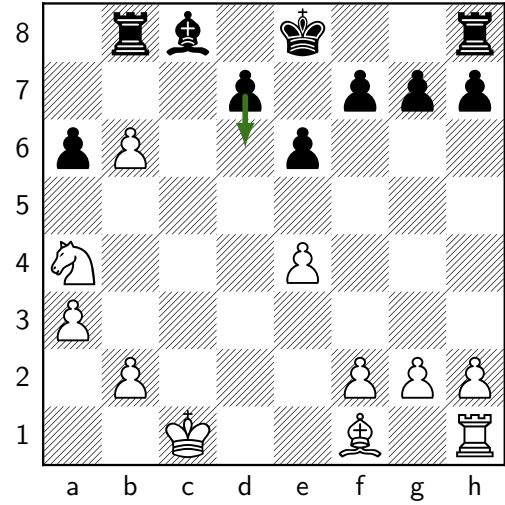
(a)



(b)



(c)



(d)

Figure 11: **Move 20, White: g3.** (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#2540` (ply 33) [structural distance $d=67$, QAP obj=4]: the master played f6; under μ our $Pg2$ plays the role of that game’s $Pf2$; cross-type roles: our $Pf3$ as their $Nc6$, our $Nc3$ as their $Pd6$; transferred move g2g3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#1099` (ply 55) [structural distance $d=51$, QAP obj=3]: the master played f6; under μ our $Pg2$ plays the role of that game’s $Pf2$; cross-type roles: our $bc8$ as their $pf3$, our $pg5$ as their $nc2$; transferred move g2g3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#421` (ply 35) [structural distance $d=52$, QAP obj=3]: the master played d6; under μ our $Pg2$ plays the role of that game’s $Pd2$; cross-type roles: our $Nc3$ as their $Rb8$, our $ne7$ as their $pb6$; transferred move g2g3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to g3 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

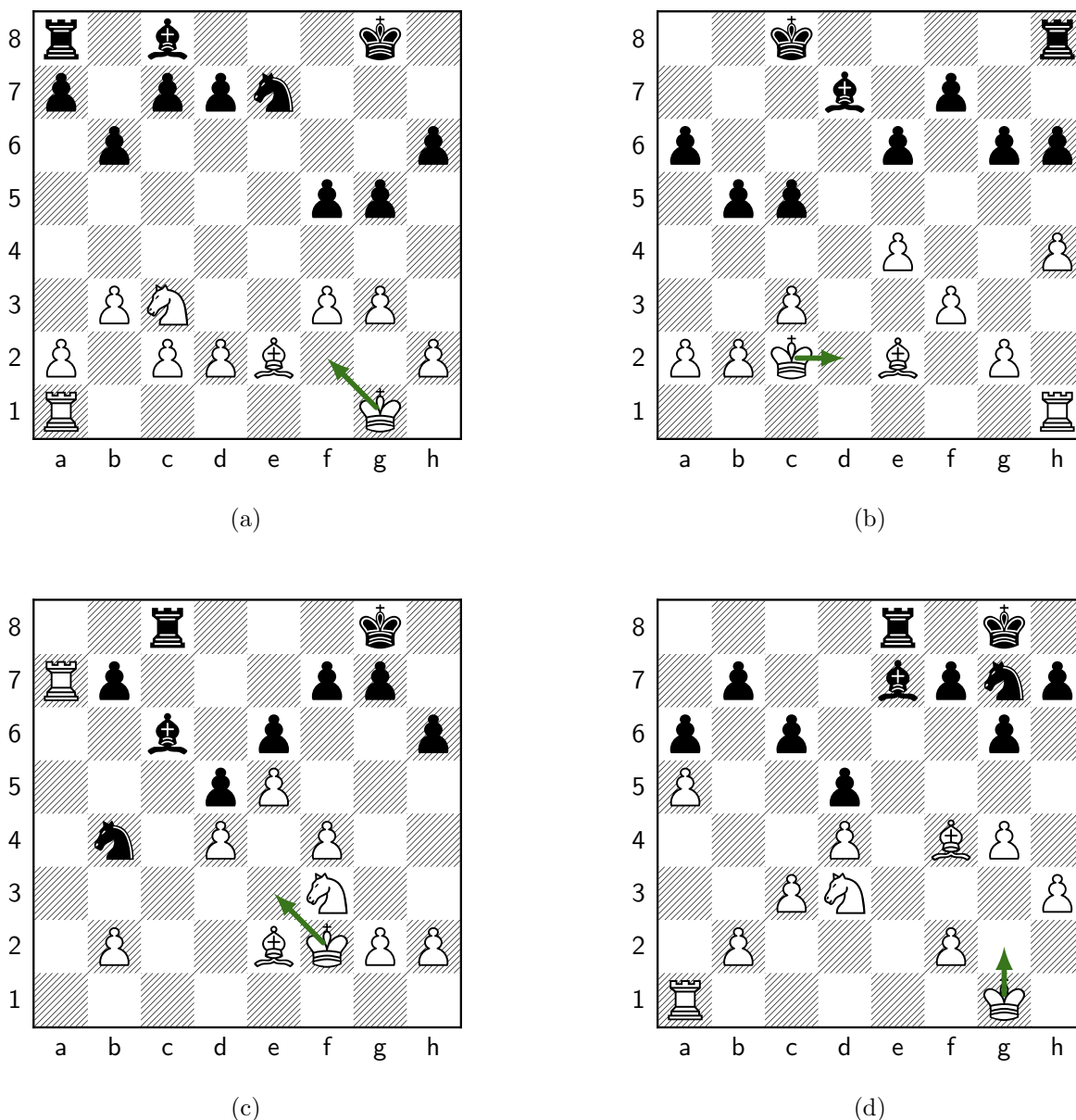
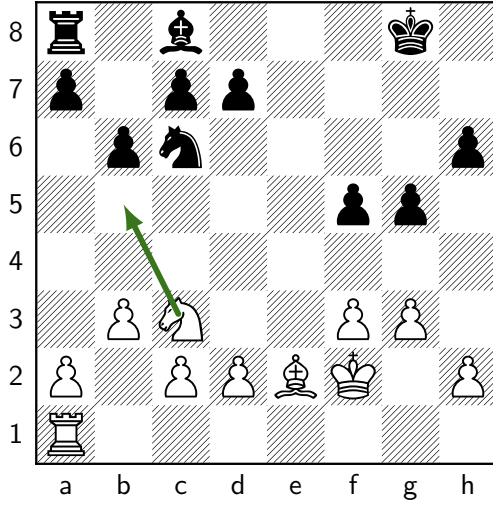
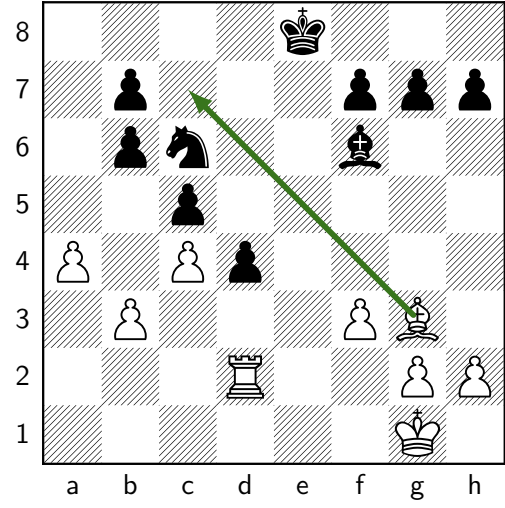


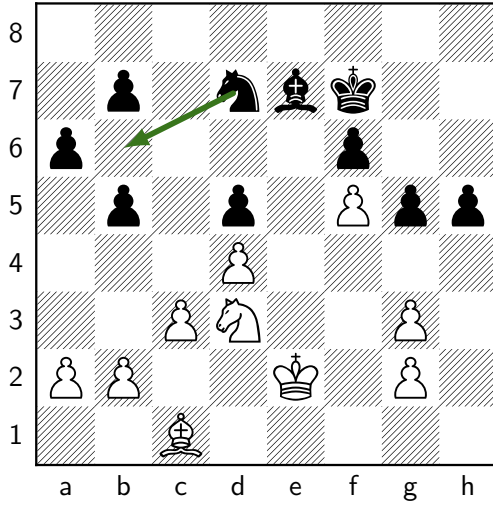
Figure 12: **Move 21, White: Kf2.** (a) Our position; Melanie plays Kf2 (g1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#3104` (ply 48) [structural distance $d=66$, QAP obj=5]: the master played Kd2; under μ our Kg1 plays the role of that game’s Kc2; cross-type roles: our pa7 as their Rh1, our Pf3 as their Be2; transferred move g1f2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#867` (ply 46) [structural distance $d=63$, QAP obj=4]: the master played Ke3; under μ our Kg1 plays the role of that game’s Kf2; cross-type roles: our Ra1 as their Pb2, our Nc3 as their Be2; transferred move g1f2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#3288` (ply 48) [structural distance $d=63$, QAP obj=4]: the master played Kg2; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nc3 as their Pc3, our pg5 as their Nd3; transferred move g1f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf2 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



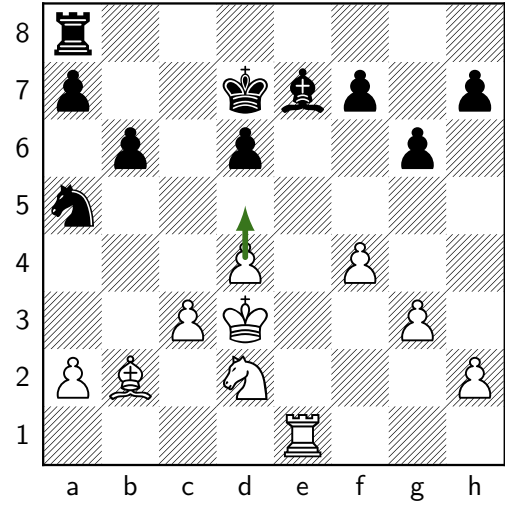
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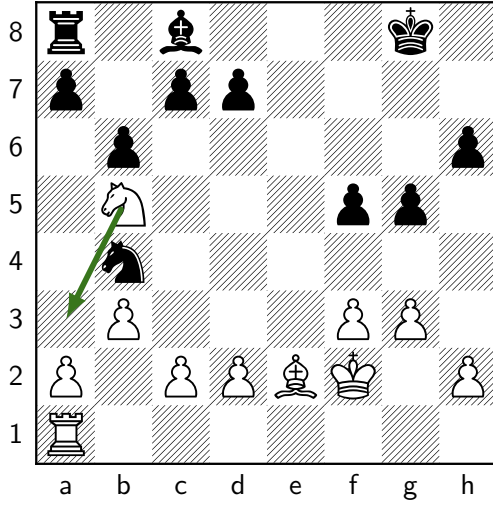


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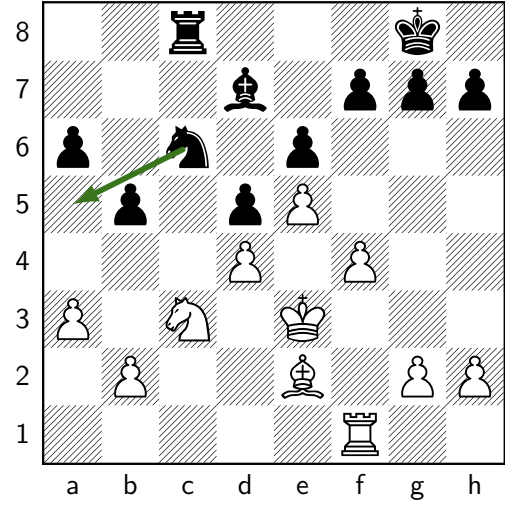


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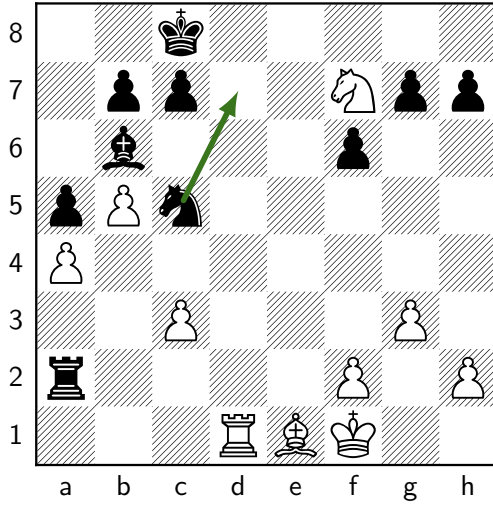
Figure 13: **Move 22, White: Nb5.** (a) Our position; Melanie plays Nb5 (c3b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#2349` (ply 58) [structural distance $d=58$, QAP obj=8]: the master played Bc7; under μ our $Nc3$ plays the role of that game’s Bg3 (a **cross-type** correspondence); cross-type roles: our Pf3 as their Rd2, our Be2 as their nc6; transferred move c3b5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#1184` (ply 59) [structural distance $d=58$, QAP obj=9]: the master played Nb6; under μ our $Nc3$ plays the role of that game’s Nd2; cross-type roles: our Ra1 as their Pb7, our nc6 as their pc3; transferred move c3a4. (d) Retrieved from `lichess_elite_2024-09_p0026#239` (ply 58) [structural distance $d=54$, QAP obj=8]: the master played d5; under μ our Pf3 plays the role of that game’s Pd4; cross-type roles: our Nc3 as their Re1, our ra8 as their Bb2; transferred move f3f4. Synthesis: 1 retrieved analog transferred to Nb5 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



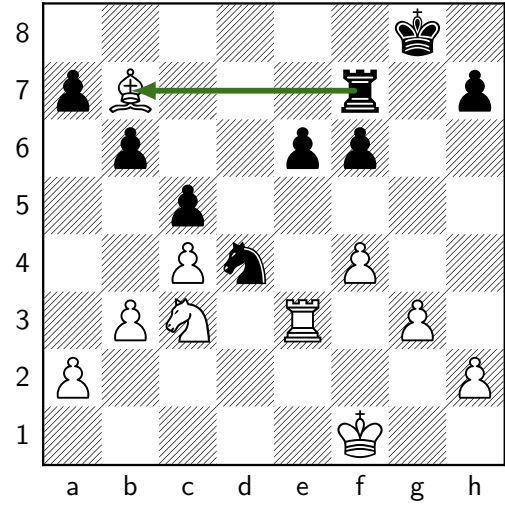
(a)



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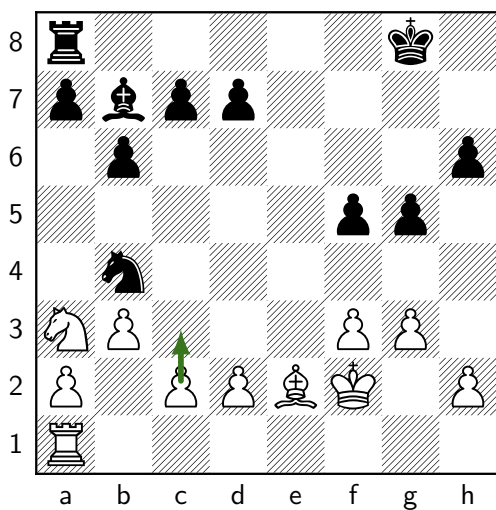


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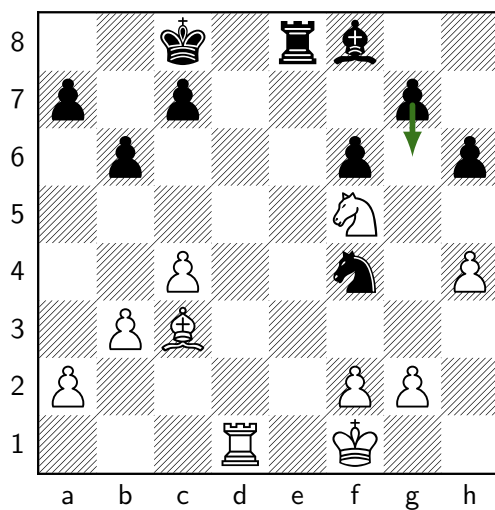


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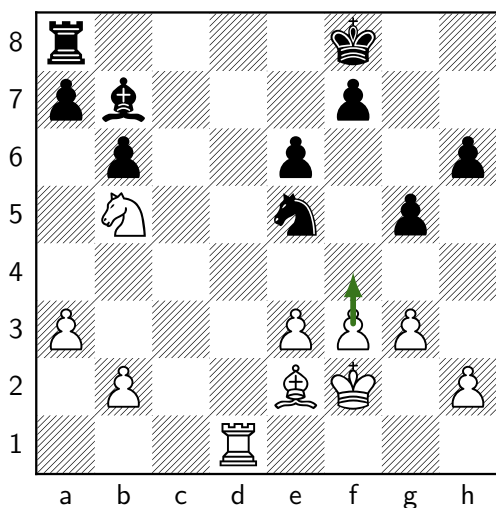
Figure 14: **Move 23, White: Na3.** (a) Our position; Melanie plays Na3 (b5a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#423` (ply 45) [structural distance $d=52$, QAP obj=100]: the master played Na5; under μ our Nb5 plays the role of that game’s Nc3; cross-type roles: our pg5 as their Rc8, our Pd2 as their Bd7; transferred move b5a3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1029` (ply 57) [structural distance $d=56$, QAP obj=47]: the master played Nd7; under μ our Nb5 plays the role of that game’s Nc4; cross-type roles: our Be2 as their Pc7, our pg5 as their Bb6; transferred move b5a3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#588` (ply 51) [structural distance $d=60$, QAP obj=32]: the master played Rxb7; under μ our Nb5 plays the role of that game’s Rf2 (a **cross-type** correspondence); cross-type roles: our pa7 as their bb7, our Ra1 as their nc3; transferred move b5a3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Na3 (summed QAP objective 215, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



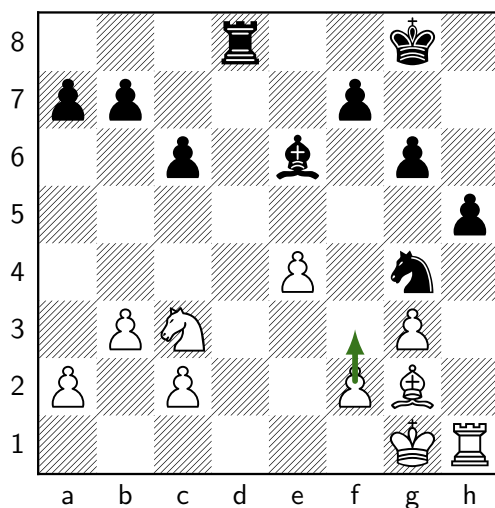
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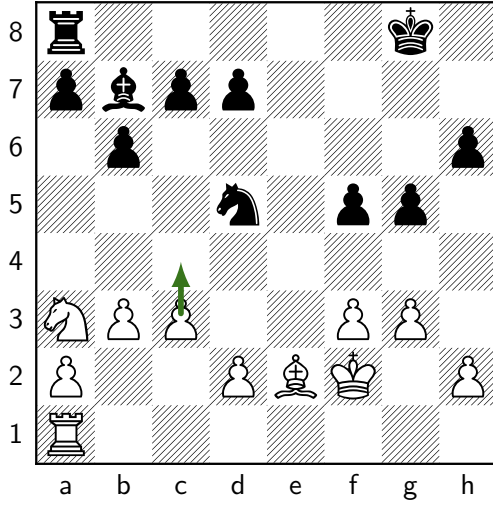


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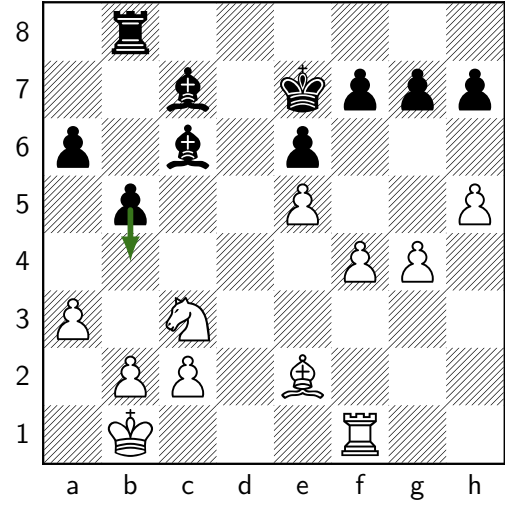


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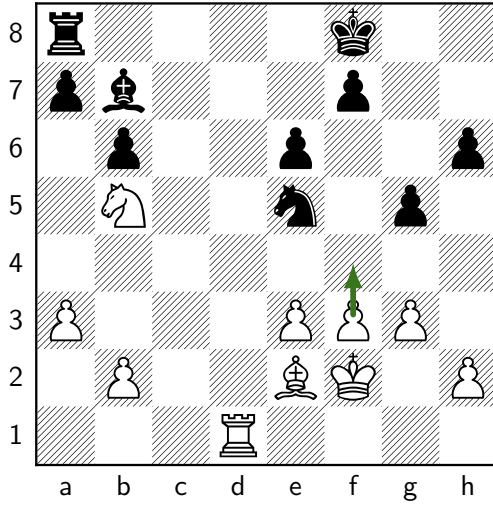
Figure 15: **Move 24, White: c3.** (a) Our position; Melanie plays c3 (c2c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#811` (ply 59) [structural distance $d=52$, QAP obj=66]: the master played g6; under μ our $Pc2$ plays the role of that game’s $Pg2$; cross-type roles: our $Na3$ as their $Re8$, our $Pb3$ as their $Bf8$; transferred move c2c3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#727` (ply 42) [structural distance $d=49$, QAP obj=44]: the master played f4; under μ our $Pc2$ plays the role of that game’s $Pf3$; cross-type roles: our $Be2$ as their $Rd1$, our $Ra1$ as their $Be2$; transferred move c2c3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#1788` (ply 48) [structural distance $d=57$, QAP obj=39]: the master played f3; under μ our $Pc2$ plays the role of that game’s $Pf2$; cross-type roles: our $bb7$ as their $Rh1$, our $Be2$ as their $Pc2$; transferred move c2c3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to c3 (summed QAP objective 202, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



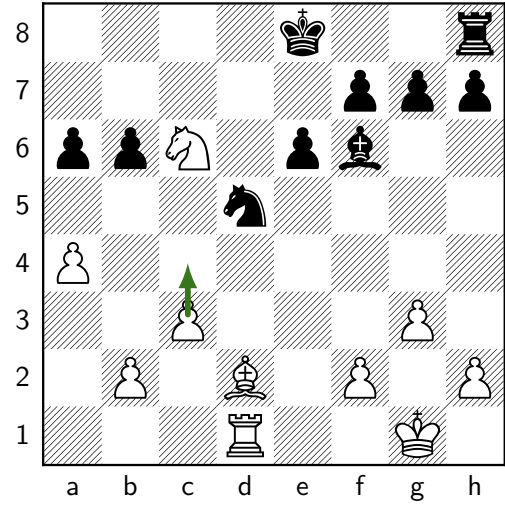
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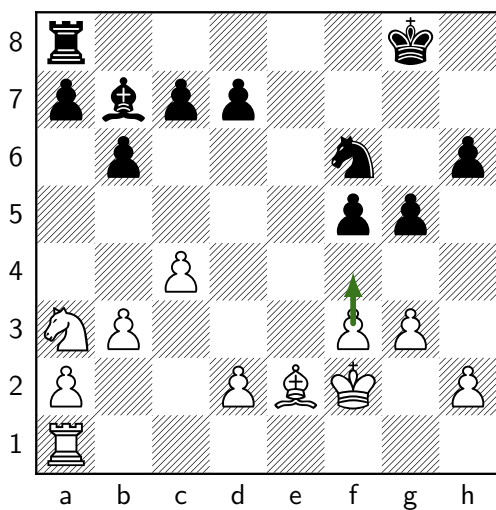


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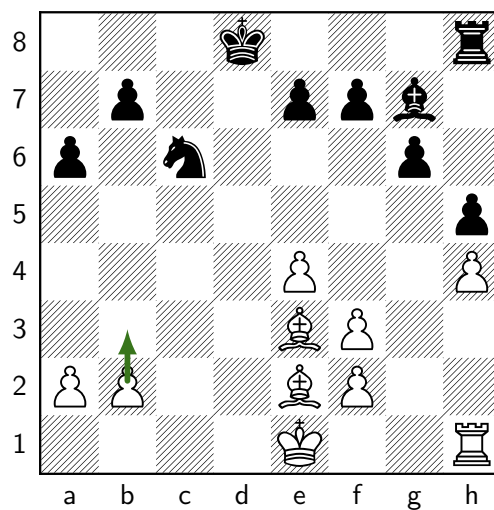


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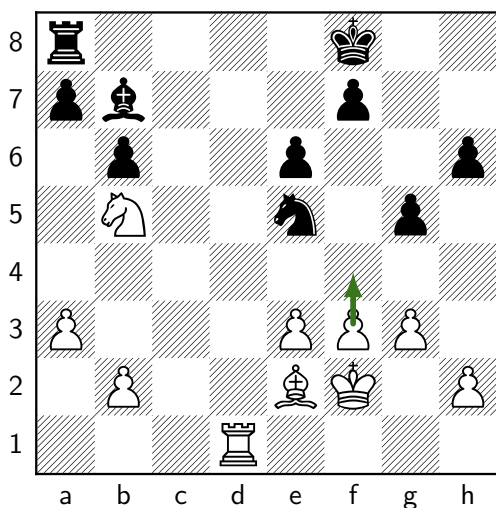
Figure 16: **Move 25, White: c4.** (a) Our position; Melanie plays c4 (c3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#2794` (ply 47) [structural distance $d=56$, QAP obj=41]: the master played b4; under μ our $Pc3$ plays the role of that game’s $Pb4$; cross-type roles: our $nd5$ as their $Bc7$, our $Pb3$ as their $Bc6$; transferred move c3c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#727` (ply 42) [structural distance $d=51$, QAP obj=40]: the master played f4; under μ our $Pc3$ plays the role of that game’s $Pf3$; cross-type roles: our $Pb3$ as their $Rd1$, our $Ra1$ as their $Pb2$; transferred move c3c4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#1529` (ply 42) [structural distance $d=55$, QAP obj=35]: the master played c4; under μ our $Pc3$ plays the role of that game’s $Pc3$; cross-type roles: our $Pf3$ as their $Rd1$, our $Pg3$ as their $Bd2$; transferred move c3c4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to c4 (summed QAP objective 206, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



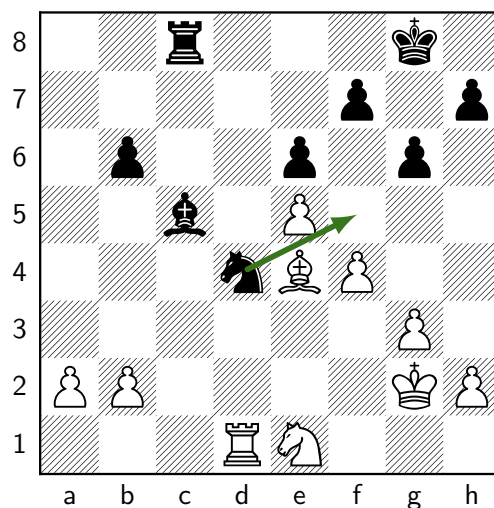
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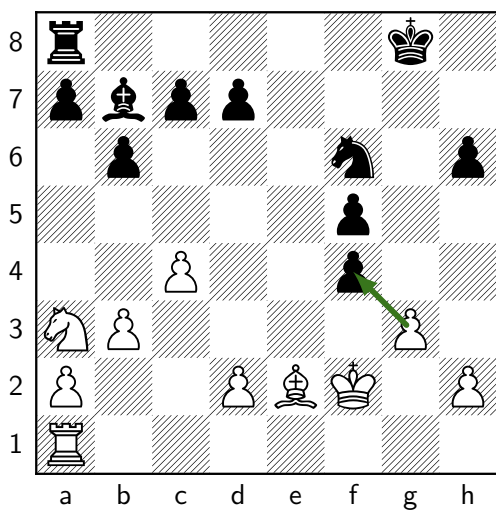


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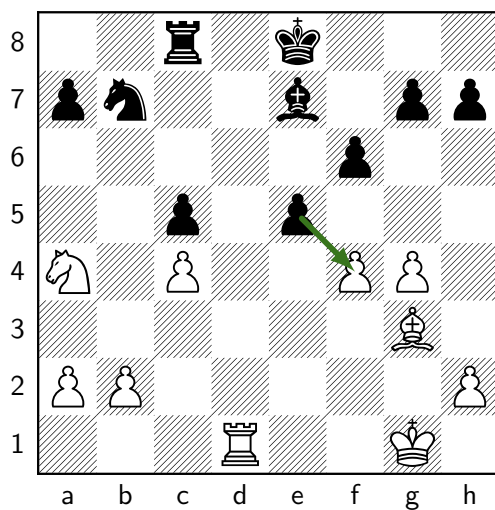


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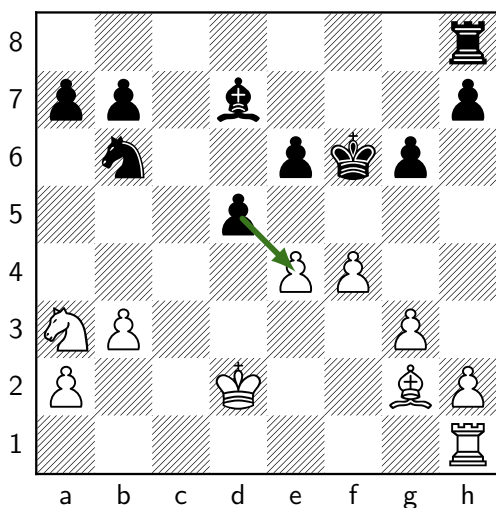
Figure 17: **Move 26, White: f4.** (a) Our position; Melanie plays f4 (f3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0020#1572 (ply 36) [structural distance $d=62$, QAP obj=31]: the master played b3; under μ our $Pf3$ plays the role of that game’s $Pb2$; cross-type roles: our $Na3$ as their $Rh1$, our $Ra1$ as their $Pa2$; transferred move f3f4 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0003#727 (ply 42) [structural distance $d=53$, QAP obj=23]: the master played f4; under μ our $Pf3$ plays the role of that game’s $Pf3$; cross-type roles: our $Pc4$ as their $Rd1$, our $Ra1$ as their $Pb2$; transferred move f3f4 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0024#2954 (ply 53) [structural distance $d=63$, QAP obj=11]: the master played Nf5; under μ our $Pf3$ plays the role of that game’s $Nd5$ (a **cross-type** correspondence); cross-type roles: our $Be2$ as their $Rc8$, our $Na3$ as their $Pg6$; transferred move f3f4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to f4 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



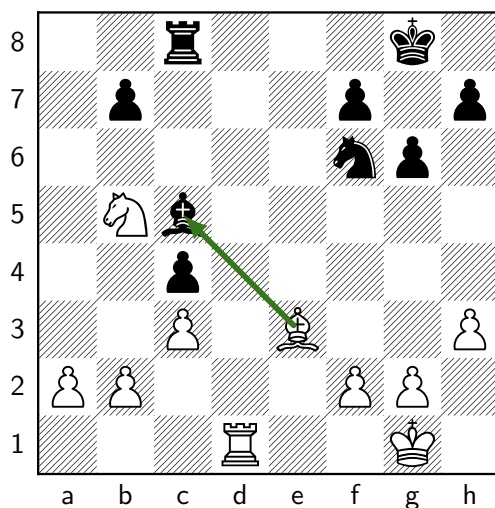
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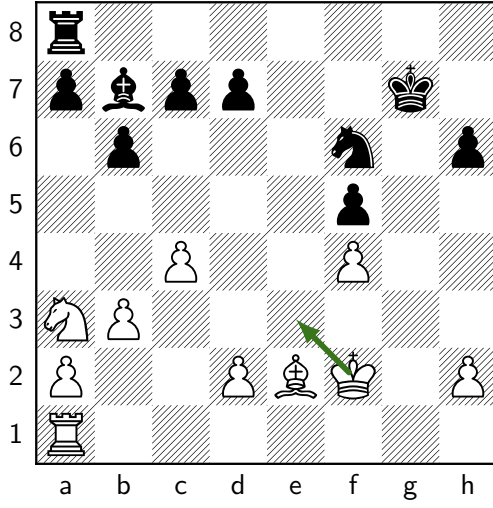


(c)

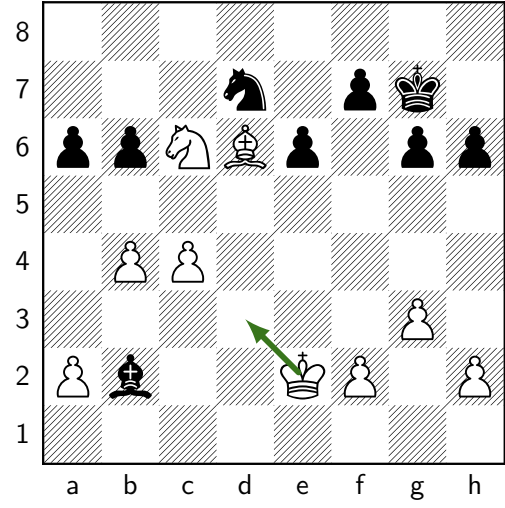


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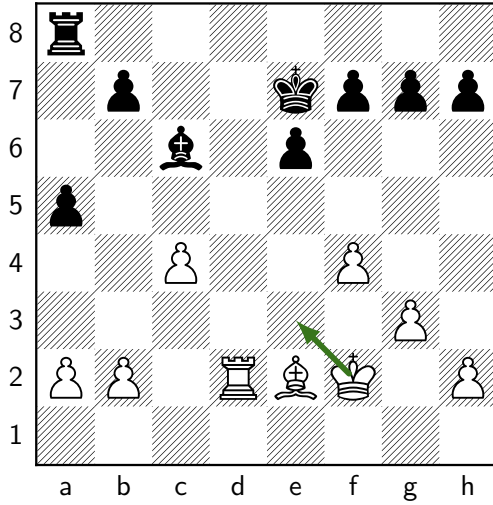
Figure 18: **Move 27, White: gxf4**. (a) Our position; Melanie plays gxf4 (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1669` (ply 53) [structural distance $d=46$, QAP obj=41]: the master played **exf4**; under μ our $Pg3$ plays the role of that game’s $Pe4$; cross-type roles: our $Be2$ as their $Rc8$, our $Pa2$ as their $Nb7$; transferred move **g3f4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#1319` (ply 45) [structural distance $d=44$, QAP obj=40]: the master played **dxe4**; under μ our $Pg3$ plays the role of that game’s $Pd4$; cross-type roles: our $Pc4$ as their $Bd7$, our $Be2$ as their $Ph7$; transferred move **g3f4** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#3345` (ply 50) [structural distance $d=52$, QAP obj=29]: the master played **Bxc5**; under μ our $Pg3$ plays the role of that game’s $Be3$ (a **cross-type** correspondence); cross-type roles: our $Be2$ as their $Rd1$, our $Ra1$ as their $Pa2$; transferred move **g3f4** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to **gxf4** (summed QAP objective 181, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



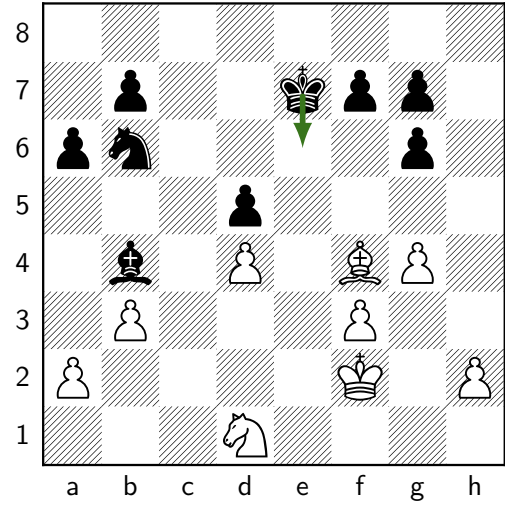
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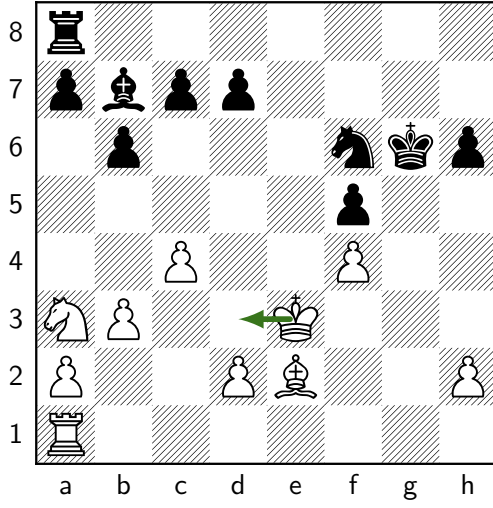


(c)

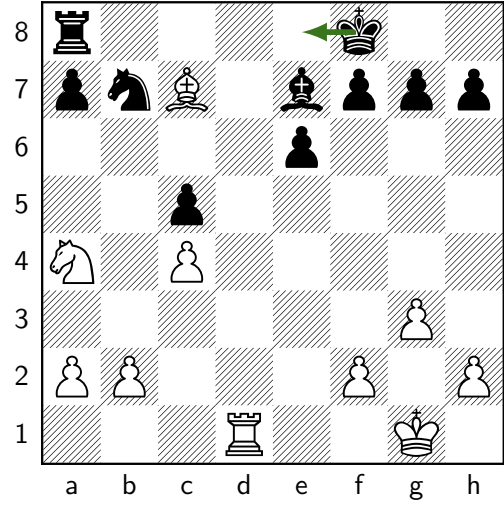


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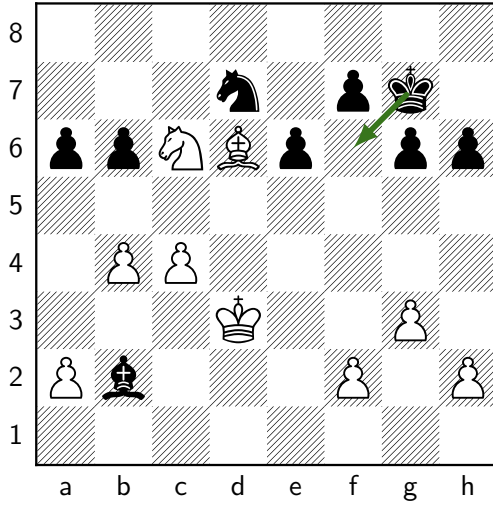
Figure 19: **Move 28, White: Ke3.** (a) Our position; Melanie plays Ke3 (f2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `twic1500#243` (ply 62) [structural distance $d=37$, QAP obj=3]: the master played Kd3; under μ our *Kf2* plays the role of that game’s *Ke2*; cross-type roles: our *Ra1* as their *Pa2*, our *ra8* as their *bb2*; transferred move f2e3 (**endorses this move**). (c) Retrieved from `lichess_elite.2024-09_p0021#1339` (ply 64) [structural distance $d=32$, QAP obj=3]: the master played Ke3; under μ our *Kf2* plays the role of that game’s *Kf2*; cross-type roles: our *ra8* as their *Pb2*, our *Pc4* as their *Rd2*; transferred move f2e3 (**endorses this move**). (d) Retrieved from `lichess_elite.2024-09_p0017#1270` (ply 57) [structural distance $d=34$, QAP obj=3]: the master played Ke6; under μ our *Kf2* plays the role of that game’s *Ke2*; cross-type roles: our *ra8* as their *Pf7*, our *Na3* as their *Pa6*; transferred move f2e3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke3 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



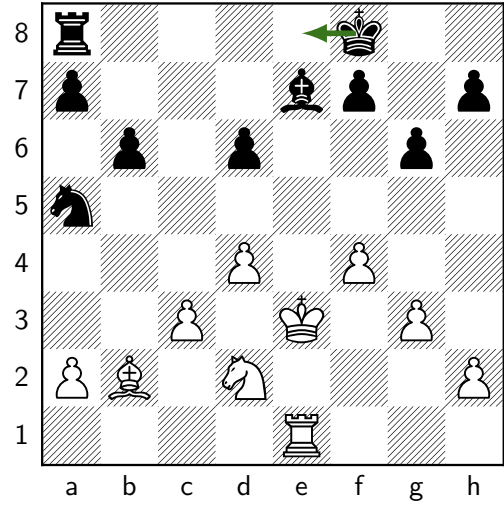
(a)



(b)



(c)



(d)

Figure 20: **Move 29, White: Kd3.** (a) Our position; Melanie plays Kd3 (e3d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1669` (ply 45) [structural distance $d=38$, QAP obj=3]: the master played Ke8; under μ our Ke3 plays the role of that game’s Kf1; cross-type roles: our pf5 as their Ra8, our Pb3 as their Nb7; transferred move e3d3 (**endorses this move**). (c) Retrieved from `twic1500#243` (ply 63) [structural distance $d=39$, QAP obj=3]: the master played Kf6; under μ our Ke3 plays the role of that game’s Kg2; cross-type roles: our pf5 as their Nd7, our ra8 as their Pf7; transferred move e3d3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#239` (ply 55) [structural distance $d=34$, QAP obj=3]: the master played Ke8; under μ our Ke3 plays the role of that game’s Kf1; cross-type roles: our Be2 as their Ra8, our Pd2 as their Be7; transferred move e3d3 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kd3 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

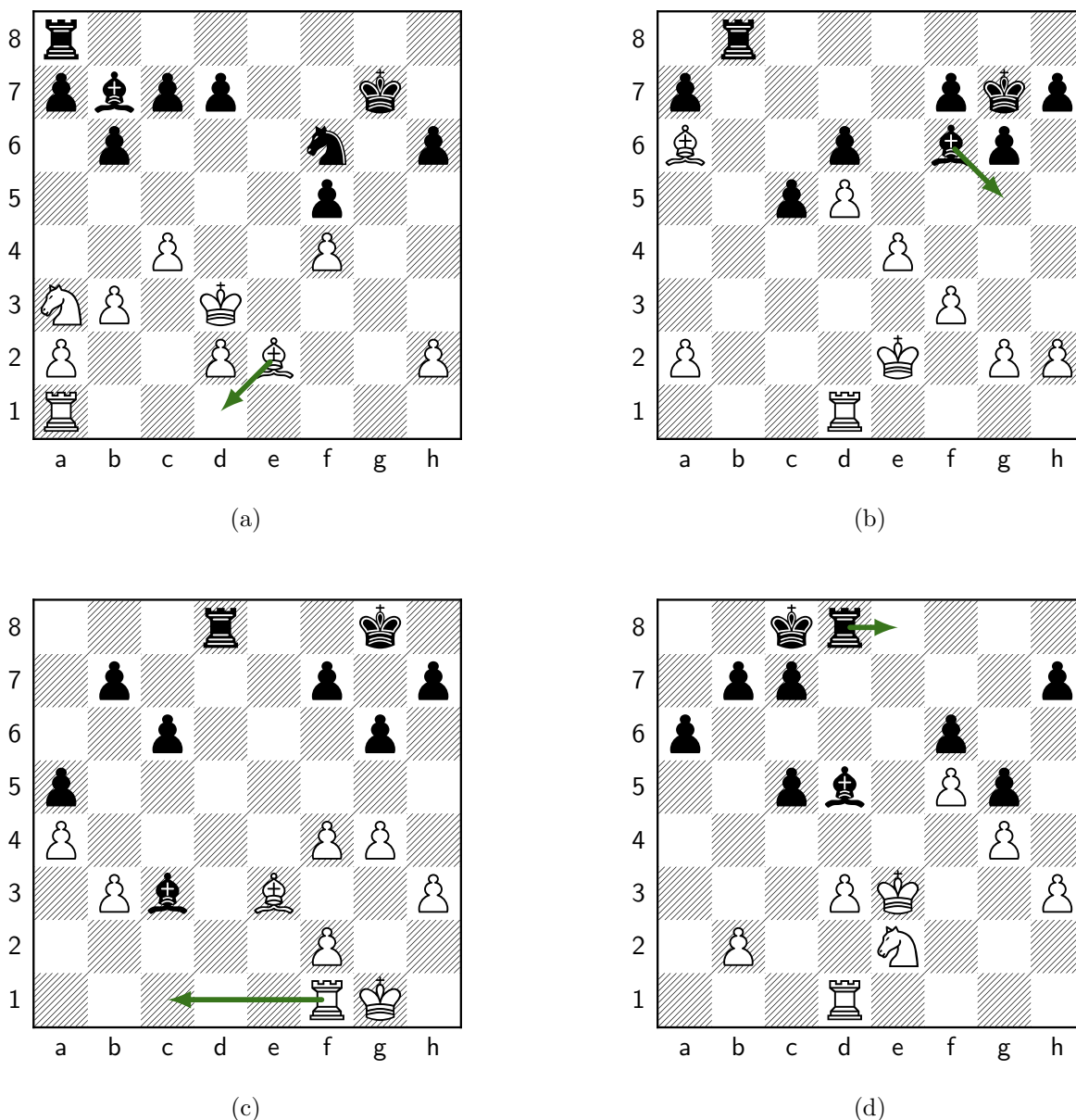
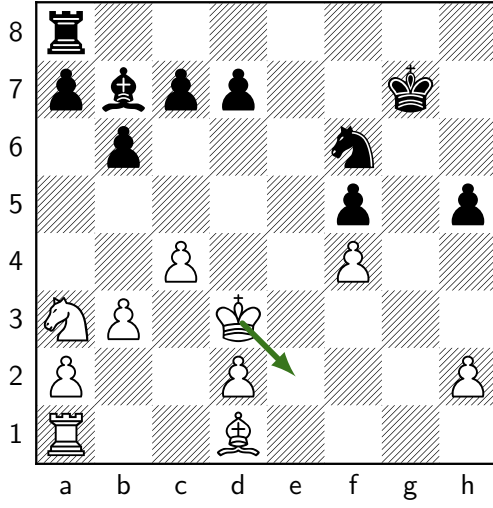
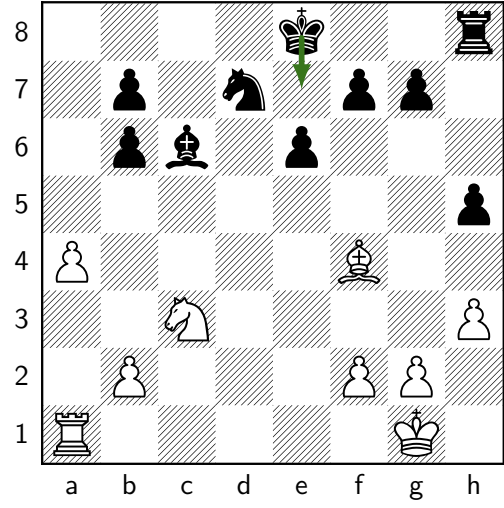


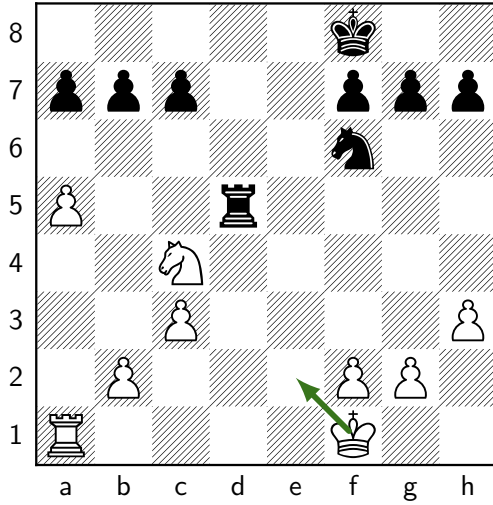
Figure 21: **Move 30, White: Bd1**. (a) Our position; Melanie plays Bd1 (e2d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#1136` (ply 51) [structural distance $d=32$, QAP obj=6]: the master played Bg5; under μ our Be2 plays the role of that game’s Bf3; cross-type roles: our pb6 as their Rb8, our bb7 as their Pf7; transferred move e2d1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#207` (ply 52) [structural distance $d=41$, QAP obj=5]: the master played Rc1; under μ our Be2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our nf6 as their bc3, our Ra1 as their Be3; transferred move e2d1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2865` (ply 57) [structural distance $d=34$, QAP obj=4]: the master played Re8+; under μ our Be2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Na3 as their Pa6, our Ra1 as their Bd5; transferred move e2d1 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Bd1 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



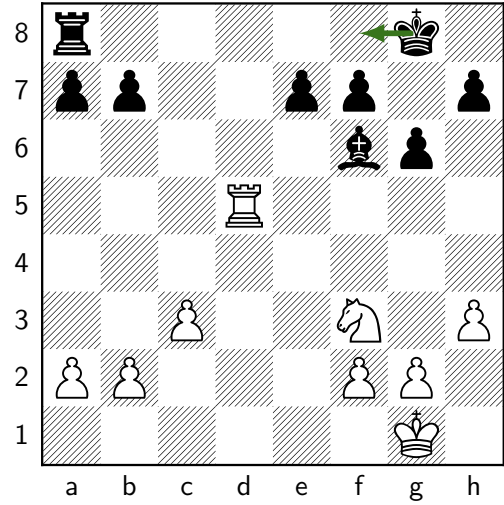
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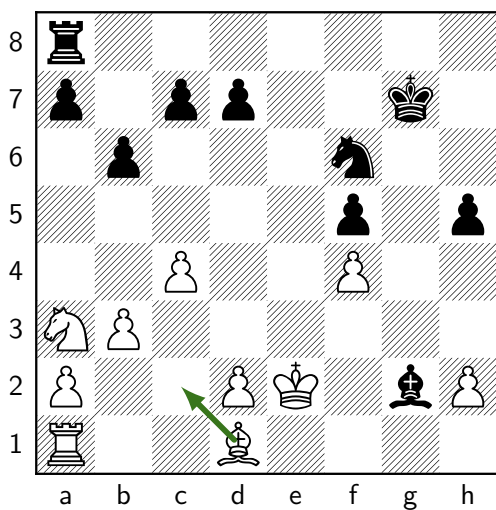


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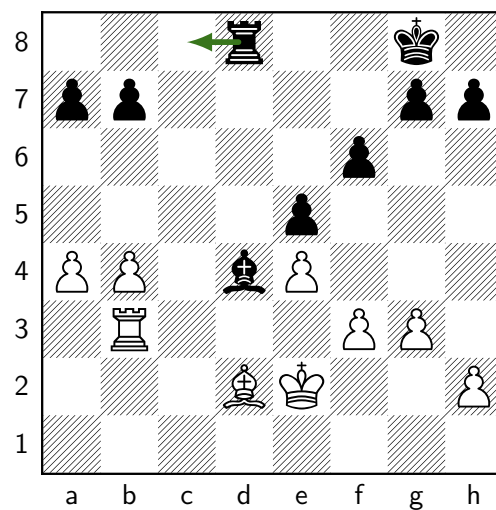


(d)

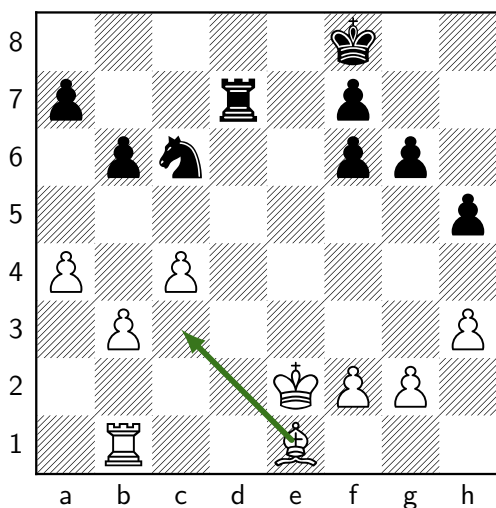
Figure 22: **Move 31, White: Ke2.** (a) Our position; Melanie plays Ke2 (d3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1620` (ply 49) [structural distance $d=36$, QAP obj=19]: the master played Ke7; under μ our *Kd3* plays the role of that game’s *Ke1*; cross-type roles: our *Pd2* as their *Rh8*, our *Pc4* as their *Nd7*; transferred move d3e2 (**endorses this move**). (c) Retrieved from `twic1500#214` (ply 52) [structural distance $d=38$, QAP obj=16]: the master played Ke2; under μ our *Kd3* plays the role of that game’s *Kf1*; cross-type roles: our *bb7* as their *Ra1*, our *ra8* as their *Ph3*; transferred move d3e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#2382` (ply 47) [structural distance $d=39$, QAP obj=16]: the master played Kf8; under μ our *Kd3* plays the role of that game’s *Kg1*; cross-type roles: our *Na3* as their *Ph7*, our *Pc4* as their *rd5*; transferred move d3e2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ke2 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



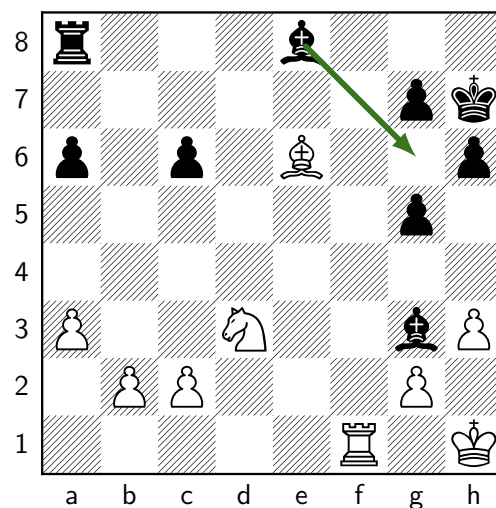
(a)



(b)

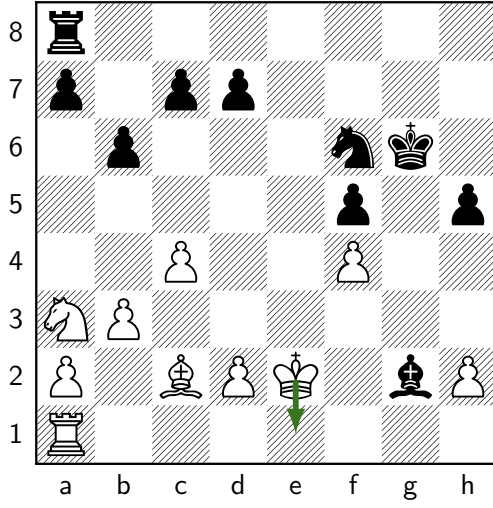


(c)

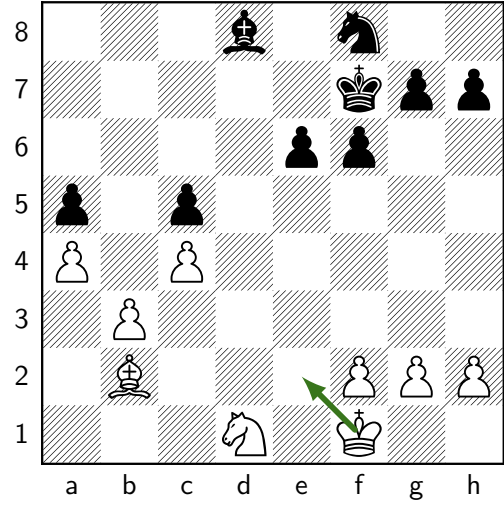


(d)

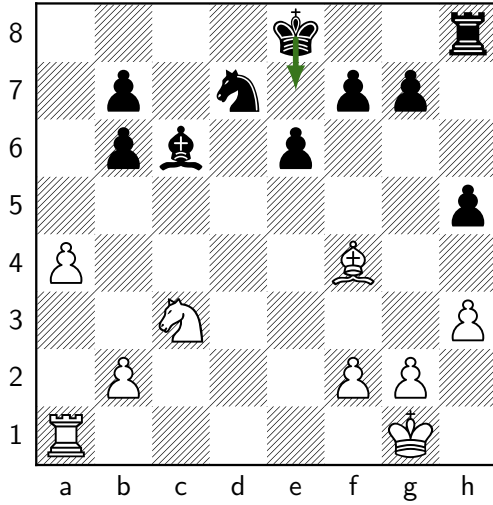
Figure 23: **Move 32, White: Bc2**. (a) Our position; Melanie plays Bc2 (d1c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#1099` (ply 69) [structural distance $d=40$, QAP obj=4]: the master played Rc8; under μ our *Bd1* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Ra1* as their *Pa7*, our *Na3* as their *Pe5*; transferred move d1c2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0023#2998` (ply 62) [structural distance $d=44$, QAP obj=4]: the master played Bc3; under μ our *Bd1* plays the role of that game’s *Be1*; cross-type roles: our *bg2* as their *rd7*, our *ra8* as their *pf7*; transferred move d1c2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#1926` (ply 63) [structural distance $d=34$, QAP obj=4]: the master played Bg6; under μ our *Bd1* plays the role of that game’s *Be1*; cross-type roles: our *nf6* as their *Ra8*, our *bg2* as their *Pg7*; transferred move d1c2 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Bc2 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



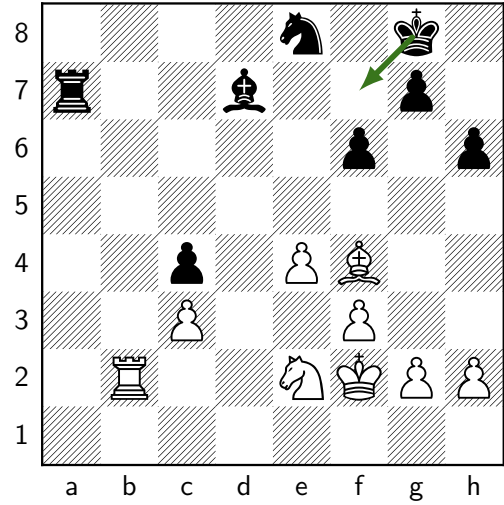
(a)



(b)

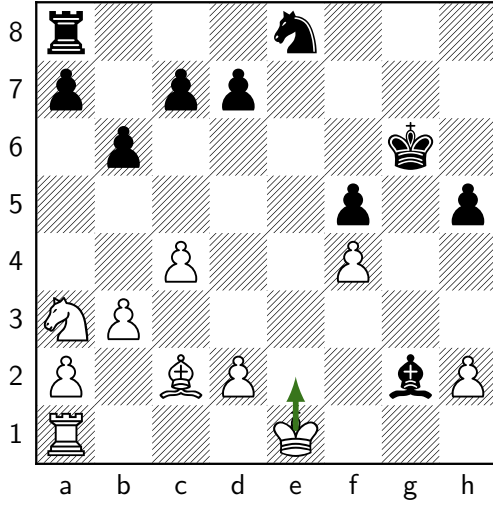


(c)

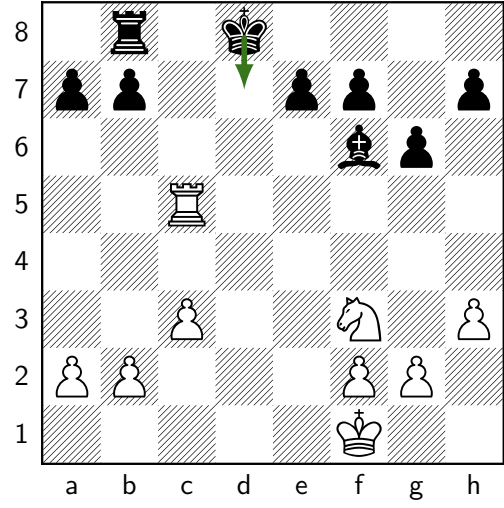


(d)

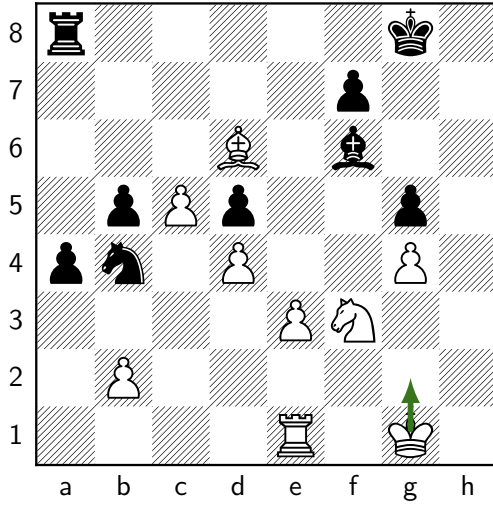
Figure 24: **Move 33, White: Ke1.** (a) Our position; Melanie plays Ke1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#1036` (ply 60) [structural distance $d=42$, QAP obj=23]: the master played Ke2; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our bg2 as their Pg2, our Ra1 as their Ph2; transferred move e2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1620` (ply 49) [structural distance $d=49$, QAP obj=22]: the master played Ke7; under μ our Ke2 plays the role of that game’s Ke1; cross-type roles: our Pf4 as their Rh8, our Na3 as their Pb7; transferred move e2e1 (**endorses this move**). (d) Retrieved from `twic1500#406` (ply 81) [structural distance $d=41$, QAP obj=20]: the master played Kf7; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our pb6 as their Ne8, our pa7 as their Bd7; transferred move e2e1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke1 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



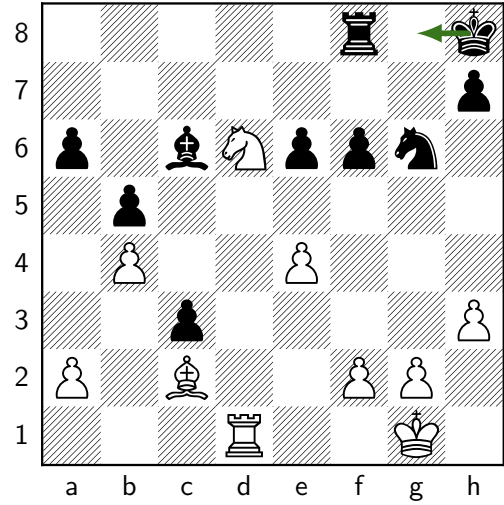
(a)



(b)

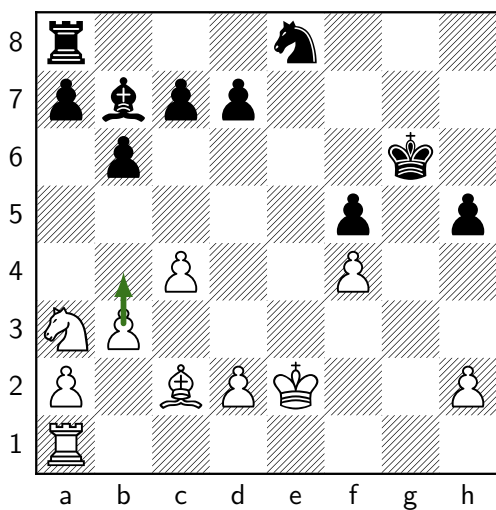


(c)

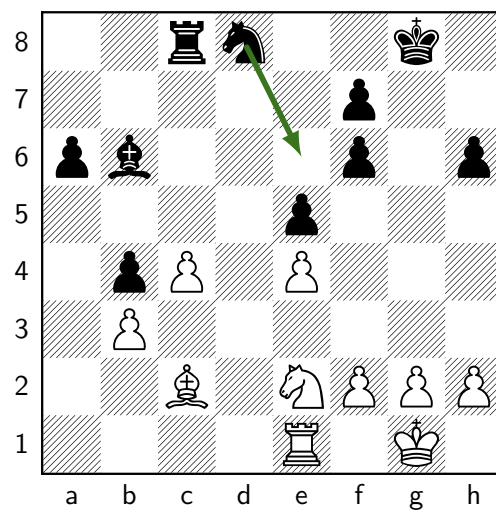


(d)

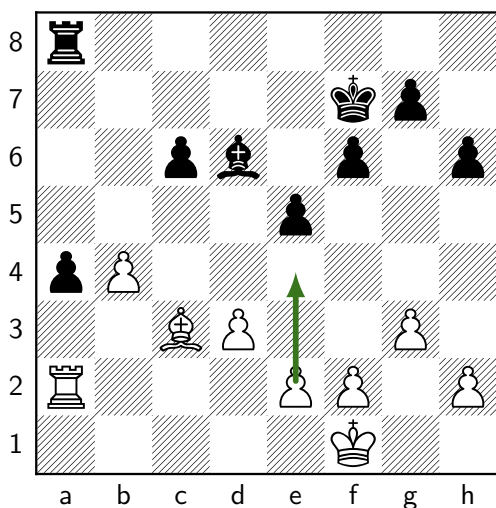
Figure 25: **Move 34, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#2382` (ply 55) [structural distance $d=47$, QAP obj=22]: the master played Kd7; under μ our Ke1 plays the role of that game’s Kd1; cross-type roles: our bg2 as their Pb7, our Na3 as their Pf7; transferred move e1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#3214` (ply 62) [structural distance $d=49$, QAP obj=18]: the master played Kg2; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our Pb3 as their Re1, our Ra1 as their Pe3; transferred move e1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#3002` (ply 59) [structural distance $d=45$, QAP obj=18]: the master played Kg8; under μ our Ke1 plays the role of that game’s Kh1; cross-type roles: our Bc2 as their nd6, our ra8 as their Ng6; transferred move e1e2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ke2 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



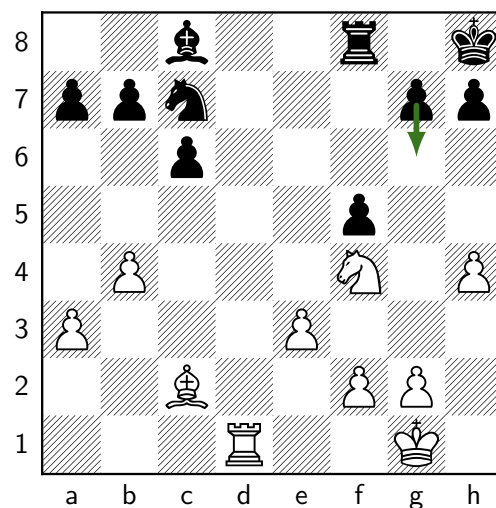
(a)



(b)

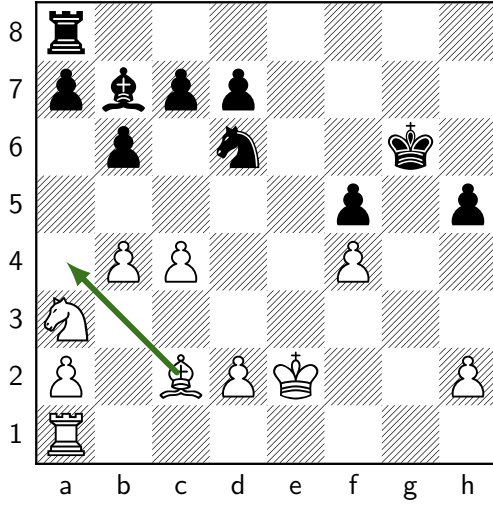


(c)

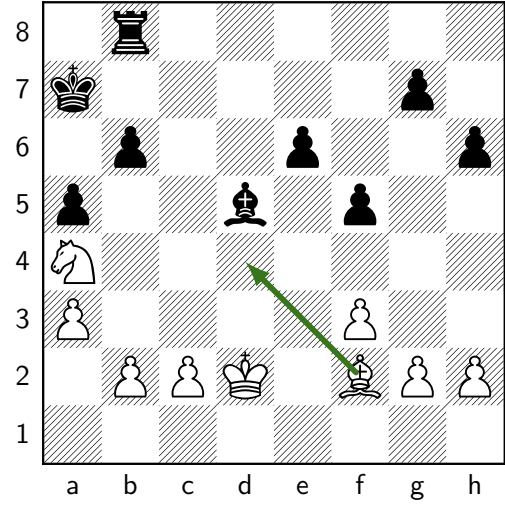


(d)

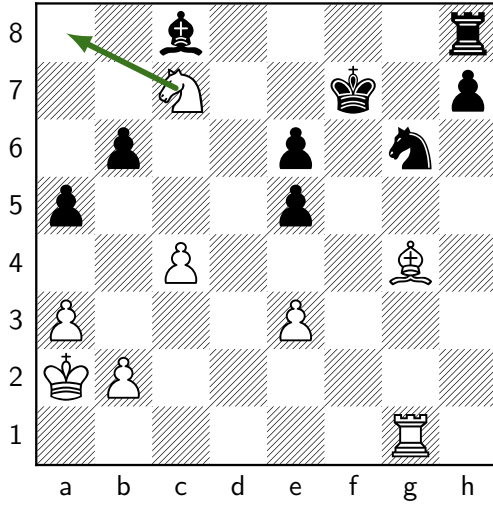
Figure 26: **Move 35, White: b4.** (a) Our position; Melanie plays b4 (b3b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1225` (ply 53) [structural distance $d=47$, QAP obj=56]: the master played Ne6; under μ our *Pb3* plays the role of that game’s *Nd1* (a **cross-type** correspondence); cross-type roles: our *Bc2* as their *Rc8*, our *Ra1* as their *Pf7*; transferred move b3b4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#3774` (ply 58) [structural distance $d=50$, QAP obj=36]: the master played e4; under μ our *Pb3* plays the role of that game’s *Pe2*; cross-type roles: our *Bc2* as their *Ra2*, our *Na3* as their *Pf2*; transferred move b3b4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0023#358` (ply 53) [structural distance $d=48$, QAP obj=20]: the master played g6; under μ our *Pb3* plays the role of that game’s *Pg2*; cross-type roles: our *ne8* as their *Bc8*, our *pc7* as their *nf4*; transferred move b3b4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to b4 (summed QAP objective 145, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



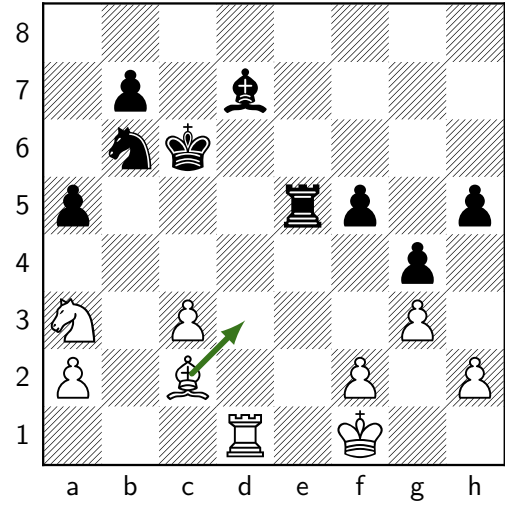
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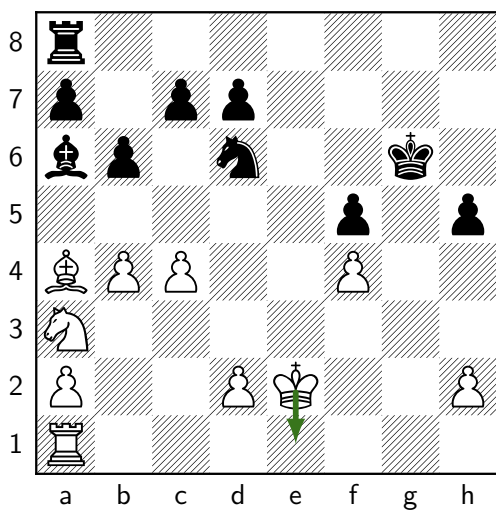


(c)

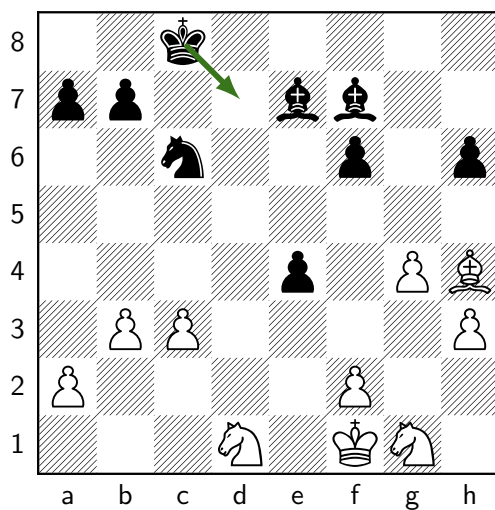


(d)

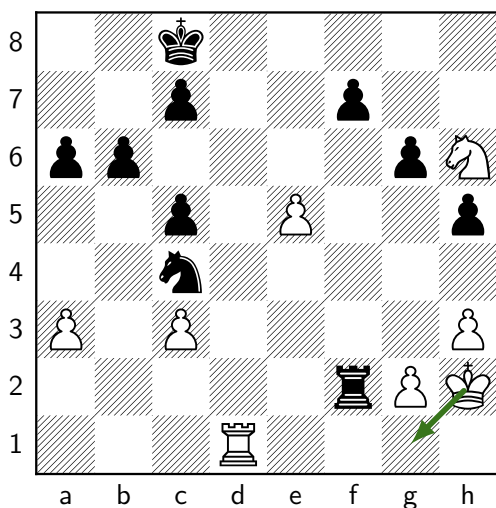
Figure 27: **Move 36, White: Ba4**. (a) Our position; Melanie plays Ba4 (c2a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3189` (ply 60) [structural distance $d=46$, QAP obj=58]: the master played Bd4; under μ our Bc2 plays the role of that game’s Bf2; cross-type roles: our Na3 as their Pg2, our Bc2 as their Na4; transferred move c2a4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#3970` (ply 58) [structural distance $d=43$, QAP obj=43]: the master played Na8; under μ our Bc2 plays the role of that game’s Nc7 (a **cross-type** correspondence); cross-type roles: our Na3 as their Rg1, our bb7 as their Pc4; transferred move c2a4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0026#565` (ply 78) [structural distance $d=53$, QAP obj=32]: the master played Bd3; under μ our Bc2 plays the role of that game’s Bc2; cross-type roles: our nd6 as their Rd1, our ph5 as their Na3; transferred move c2a4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ba4 (summed QAP objective 187, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



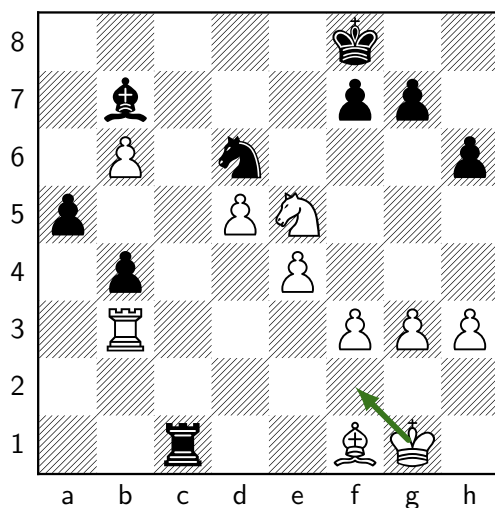
(a)



(b)

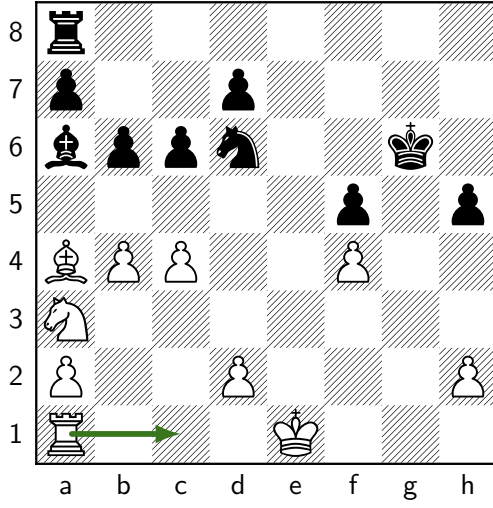


(c)

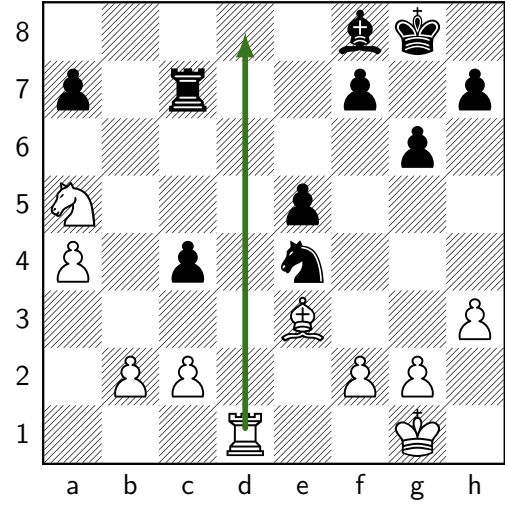


(d)

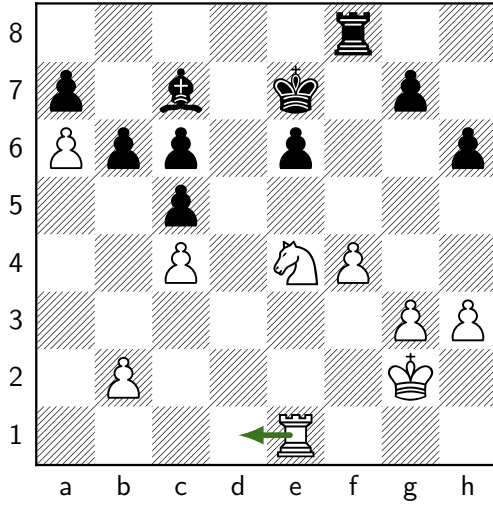
Figure 28: **Move 37, White: Ke1**. (a) Our position; Melanie plays Ke1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1506` (ply 63) [structural distance $d=50$, QAP obj=64]: the master played Kd7; under μ our Ke2 plays the role of that game’s Kc1; cross-type roles: our Na3 as their Be7, our ra8 as their Nc6; transferred move e2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#1006` (ply 82) [structural distance $d=54$, QAP obj=45]: the master played Kg1; under μ our Ke2 plays the role of that game’s Kh2; cross-type roles: our Na3 as their Rd1, our ba6 as their rf2; transferred move e2e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3305` (ply 74) [structural distance $d=54$, QAP obj=36]: the master played Kf2; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our ba6 as their rc1, our Pc4 as their Bf1; transferred move e2e1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke1 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



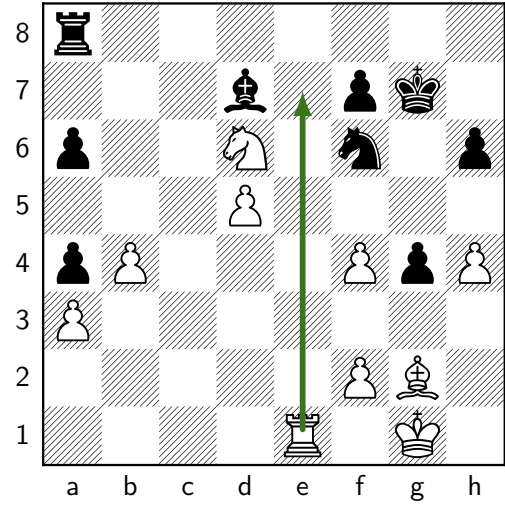
(a)



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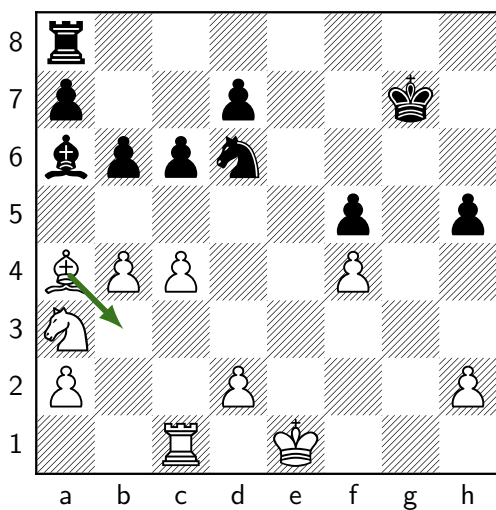


(c)

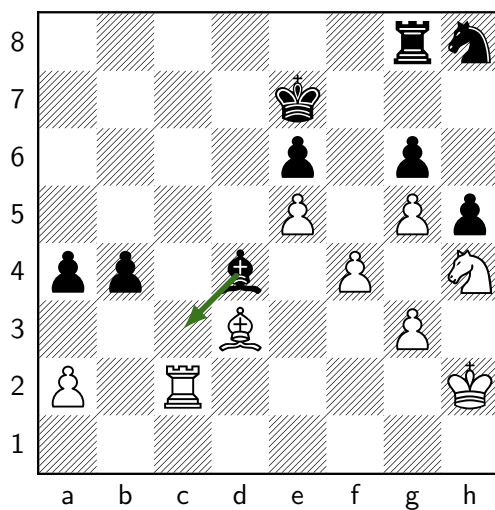


(d)

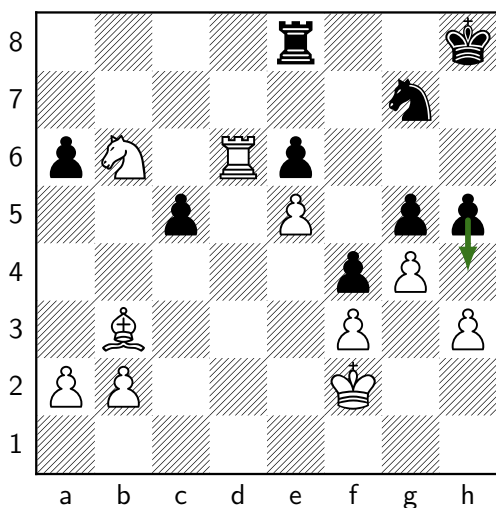
Figure 29: **Move 38, White: Rc1.** (a) Our position; Melanie plays Rc1 (a1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#3040` (ply 58) [structural distance $d=47$, QAP obj=47]: the master played Rd8; under μ our Ra1 plays the role of that game’s Rd1; cross-type roles: our Na3 as their Pc2, our ba6 as their Na5; transferred move a1c1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#443` (ply 64) [structural distance $d=44$, QAP obj=46]: the master played Rd1; under μ our Ra1 plays the role of that game’s Re1; cross-type roles: our Na3 as their Pg3, our Ba4 as their Ne4; transferred move a1c1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0016#1035` (ply 62) [structural distance $d=44$, QAP obj=43]: the master played Re7; under μ our Ra1 plays the role of that game’s Re1; cross-type roles: our Na3 as their Bg2, our Ba4 as their Nd6; transferred move a1c1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Rc1 (summed QAP objective 252, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



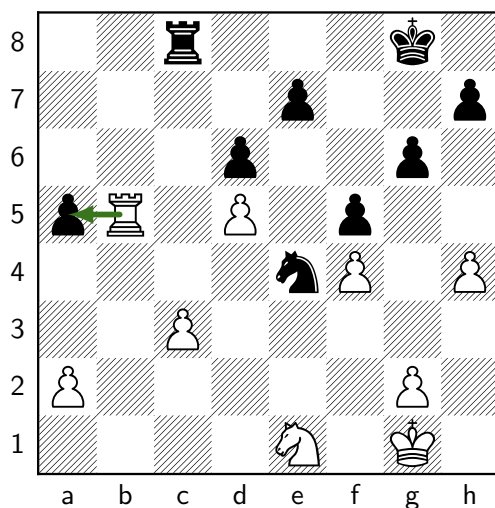
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(b)

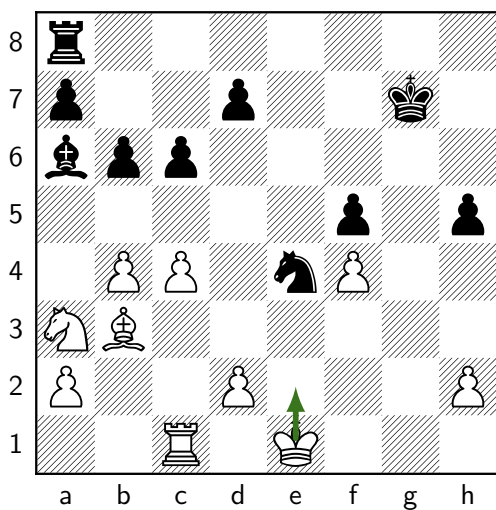


(c)

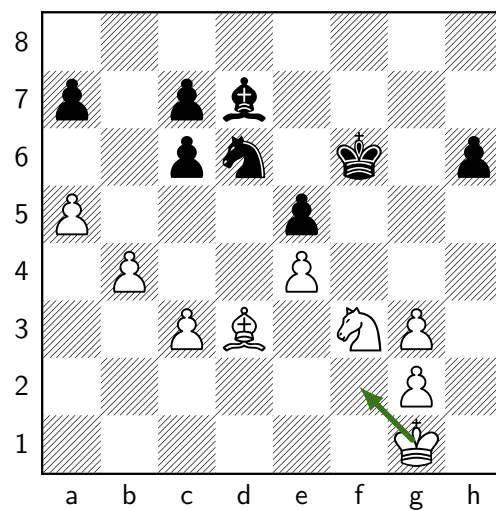


(d)

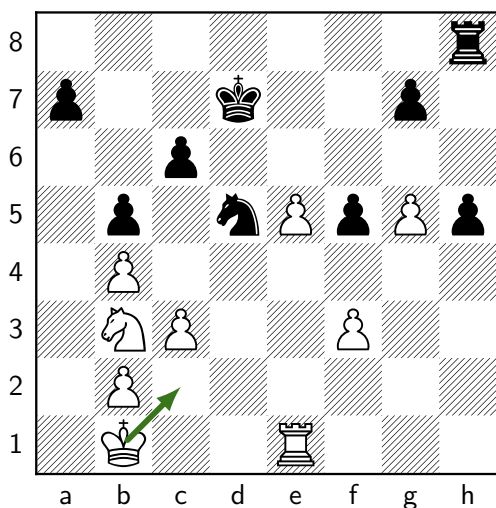
Figure 30: **Move 39, White: Bb3**. (a) Our position; Melanie plays Bb3 (a4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#1892` (ply 83) [structural distance $d=43$, QAP obj=36]: the master played Bc3; under μ our Ba4 plays the role of that game’s Bd5; cross-type roles: our Na3 as their Rg8, our Rc1 as their Nh8; transferred move a4b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#753` (ply 73) [structural distance $d=56$, QAP obj=33]: the master played h4; under μ our Ba4 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our ph5 as their nb6, our ba6 as their rd6; transferred move a4b3 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3700` (ply 68) [structural distance $d=47$, QAP obj=23]: the master played Rxa5; under μ our Ba4 plays the role of that game’s Rb5 (a **cross-type** correspondence); cross-type roles: our Rc1 as their Ne1, our ba6 as their ne4; transferred move a4b3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bb3 (summed QAP objective 153, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



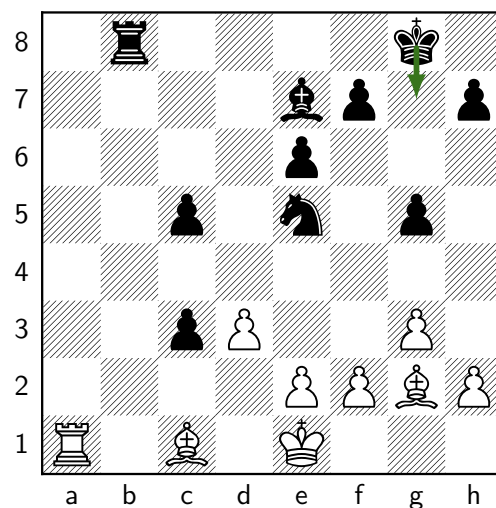
(a)



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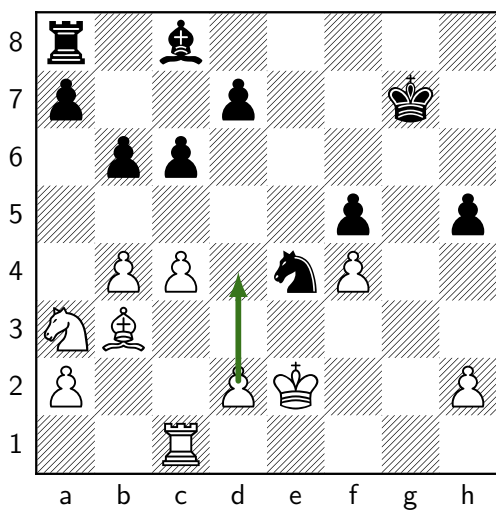


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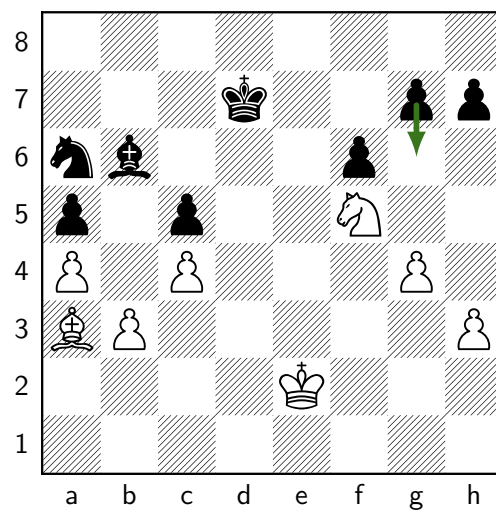


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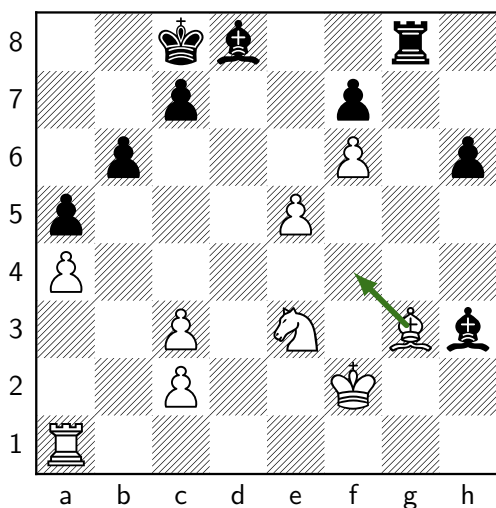
Figure 31: **Move 40, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0013#2041` (ply 72) [structural distance $d=55$, QAP obj=44]: the master played Kf2; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our Na3 as their Bd3, our Bb3 as their Pa5; transferred move e1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#2123` (ply 62) [structural distance $d=44$, QAP obj=38]: the master played Kc2; under μ our Ke1 plays the role of that game’s Kb1; cross-type roles: our Ph2 as their Nb3, our Na3 as their Pf3; transferred move e1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#2441` (ply 65) [structural distance $d=53$, QAP obj=32]: the master played Kg7; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our Pf4 as their Be7, our Bb3 as their Pc5; transferred move e1e2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Ke2 (summed QAP objective 202, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



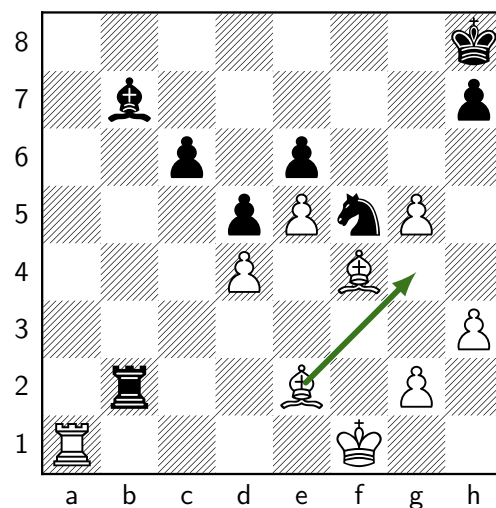
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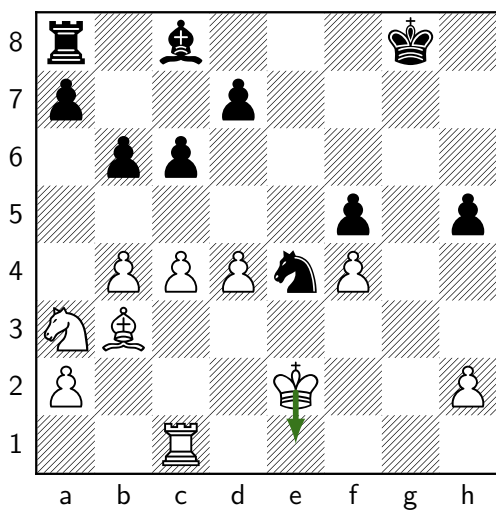


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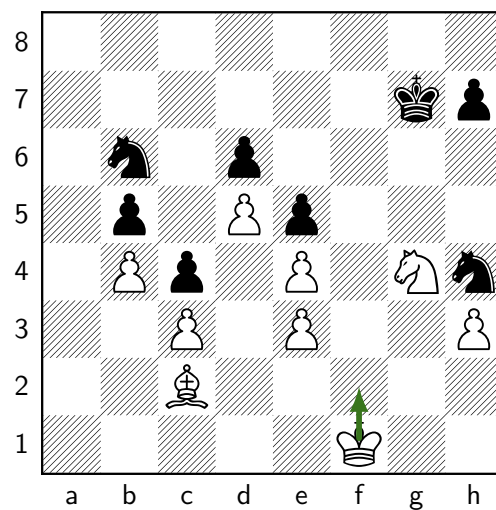


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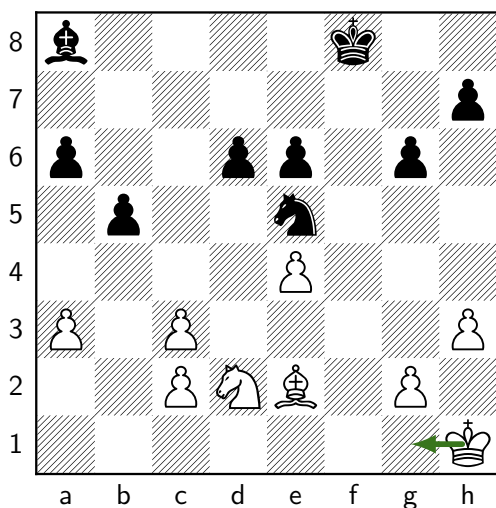
Figure 32: **Move 41, White: d4.** (a) Our position; Melanie plays d4 (d2d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#1036` (ply 83) [structural distance $d=55$, QAP obj=34]: the master played g6; under μ our $Pd2$ plays the role of that game’s $Pg2$; cross-type roles: our $Ph2$ as their $Na6$, our $pc6$ as their $Bb6$; transferred move d2d4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#1315` (ply 74) [structural distance $d=55$, QAP obj=26]: the master played Bf4; under μ our $Pd2$ plays the role of that game’s $Bg3$ (a **cross-type** correspondence); cross-type roles: our $Na3$ as their $Ra1$, our $Rc1$ as their $Pc2$; transferred move d2d4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#2906` (ply 76) [structural distance $d=49$, QAP obj=14]: the master played Bg4; under μ our $Pd2$ plays the role of that game’s $Be2$ (a **cross-type** correspondence); cross-type roles: our $Bb3$ as their $Ra1$, our $ne4$ as their $rb2$; transferred move d2d4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to d4 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



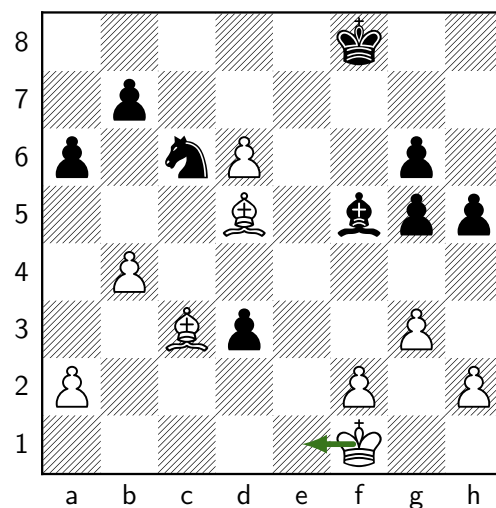
(a)



(b)

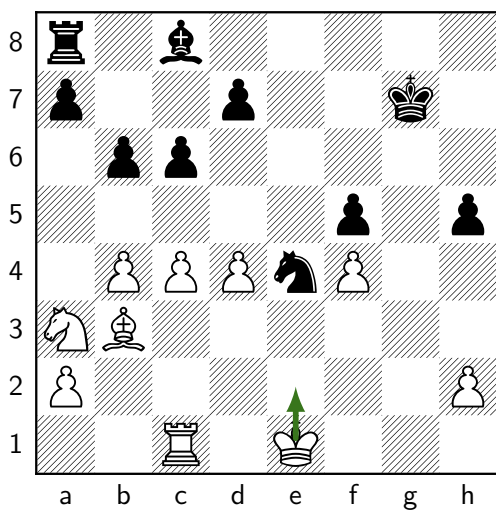


(c)

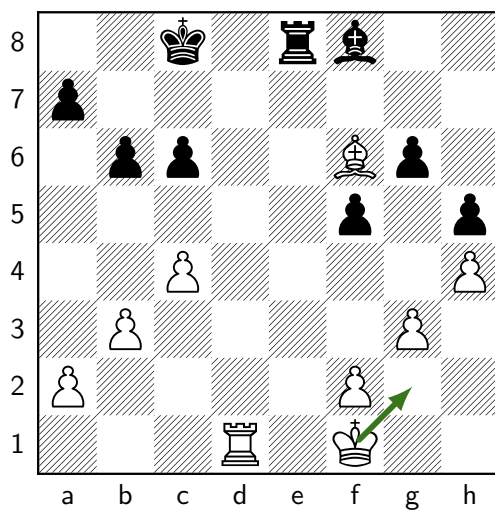


(d)

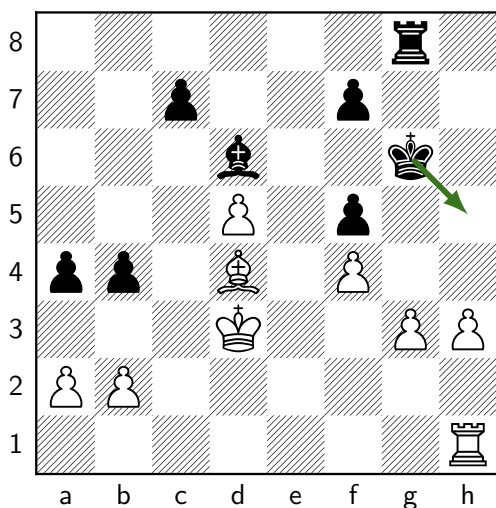
Figure 33: **Move 42, White: Ke1**. (a) Our position; Melanie plays Ke1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#3081` (ply 78) [structural distance $d=58$, QAP obj=6]: the master played Kf2; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our Pc4 as their Bc2, our bc8 as their Pb4; transferred move e2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#2661` (ply 72) [structural distance $d=54$, QAP obj=5]: the master played Kg1; under μ our Ke2 plays the role of that game’s Kh1; cross-type roles: our bc8 as their Pc2, our pb6 as their Be2; transferred move e2e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#1552` (ply 76) [structural distance $d=58$, QAP obj=5]: the master played Ke1; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our bc8 as their Pa2, our ra8 as their pd3; transferred move e2e1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ke1 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



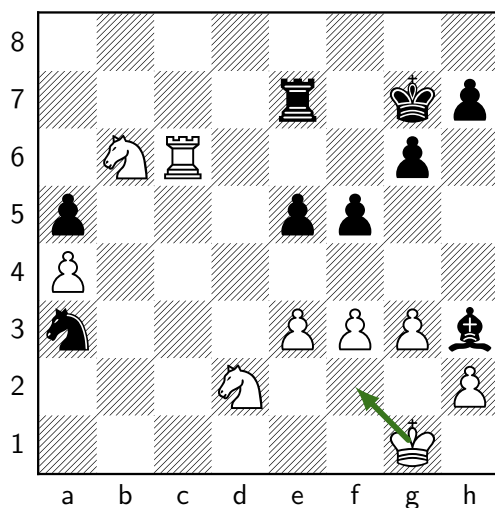
(a)



(b)



(c)



(d)

Figure 34: **Move 43, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#811` (ply 70) [structural distance $d=60$, QAP obj=6]: the master played Kg2; under μ our Ke1 plays the role of that game’s Kf1; cross-type roles: our Bb3 as their Pb3, our Pd4 as their Bf6; transferred move e1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#1514` (ply 85) [structural distance $d=57$, QAP obj=6]: the master played Kh5; under μ our Ke1 plays the role of that game’s Kg3; cross-type roles: our Pc4 as their Bd6, our Bb3 as their Pa4; transferred move e1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#2922` (ply 80) [structural distance $d=60$, QAP obj=5]: the master played Kf2; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our pc6 as their na3, our ne4 as their bh3; transferred move e1e2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ke2 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

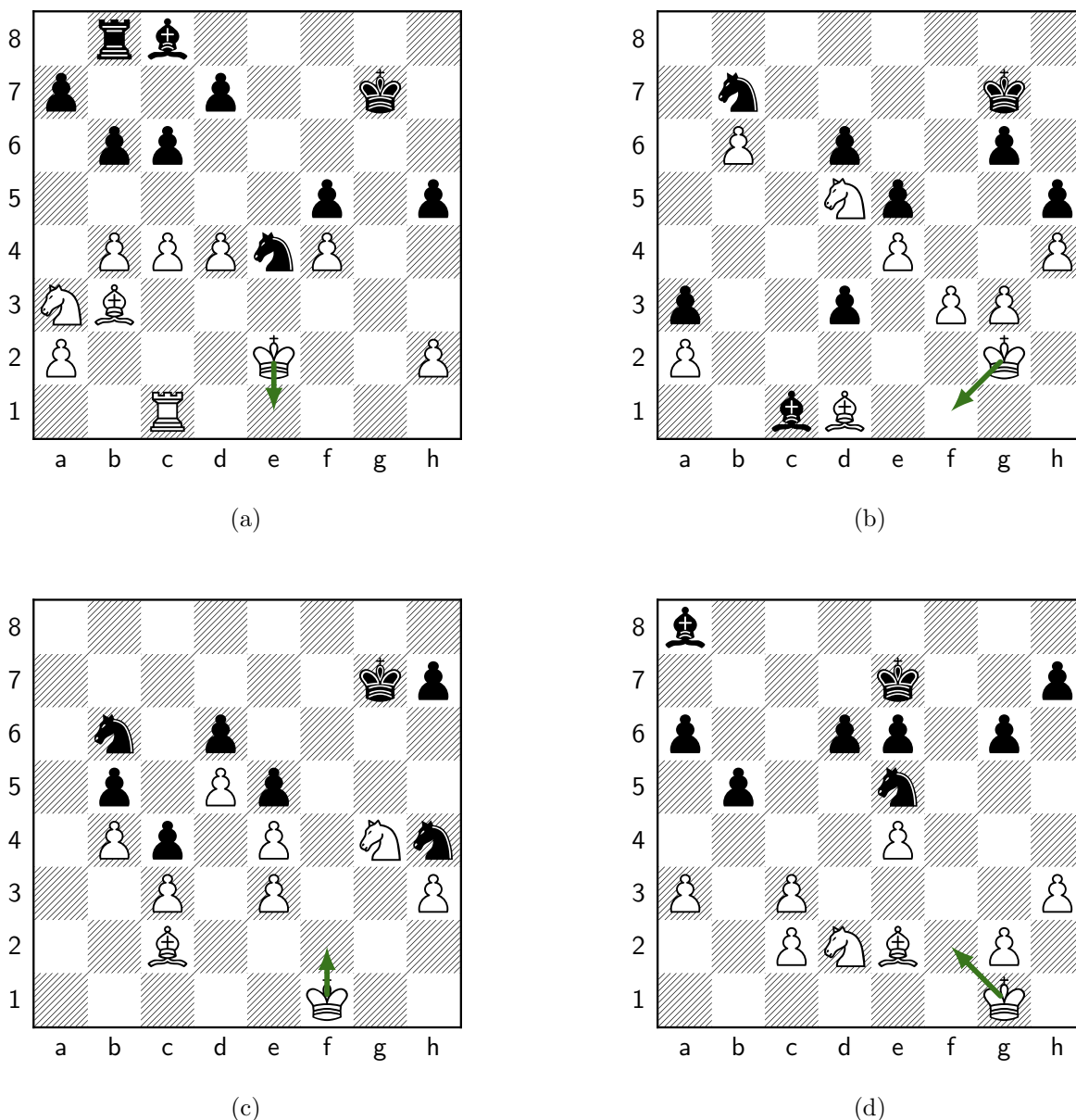
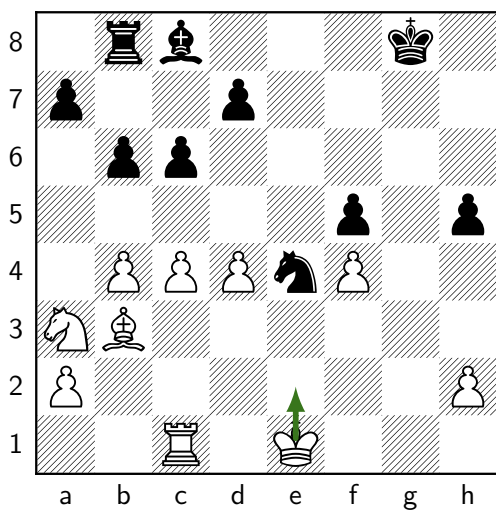
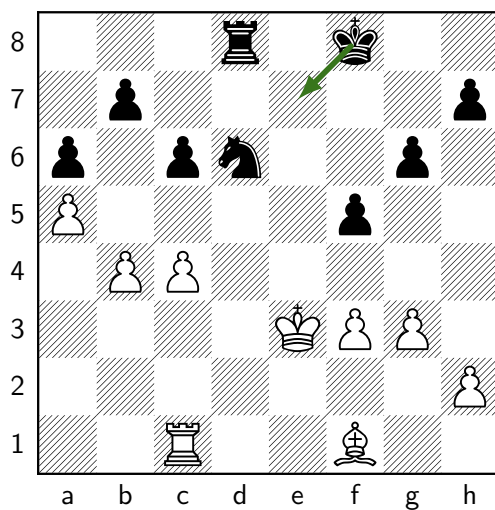


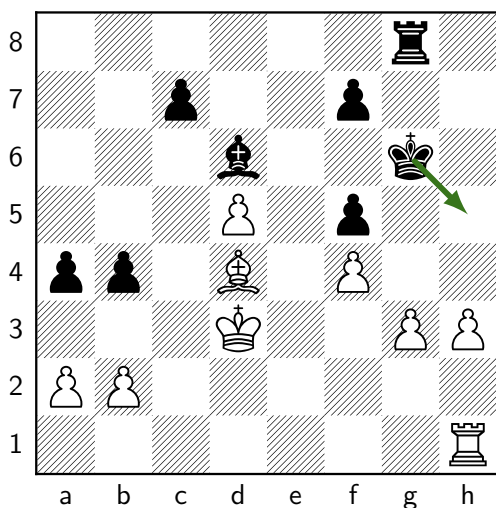
Figure 35: **Move 44, White: Ke1**. (a) Our position; Melanie plays Ke1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2354` (ply 86) [structural distance $d=62$, QAP obj=6]: the master played Kf1; under μ our Ke2 plays the role of that game’s Kg2; cross-type roles: our ne4 as their bc1, our Na3 as their Bd1; transferred move e2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#3081` (ply 78) [structural distance $d=58$, QAP obj=6]: the master played Kf2; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our Pc4 as their Bc2, our bc8 as their Pb4; transferred move e2e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#2661` (ply 74) [structural distance $d=62$, QAP obj=5]: the master played Kf2; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our Pd4 as their Nd2, our Pb4 as their Be2; transferred move e2e1 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Ke1 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



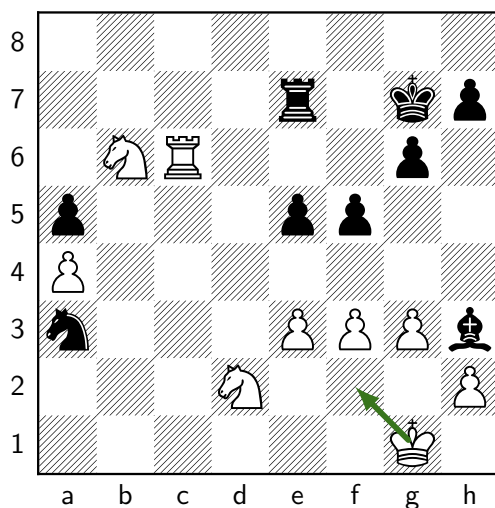
(a)



(b)

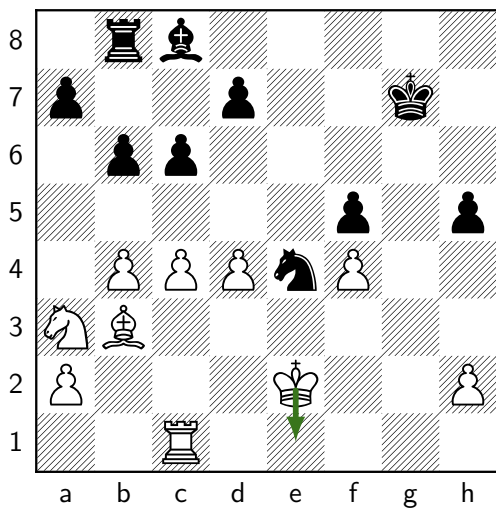


(c)

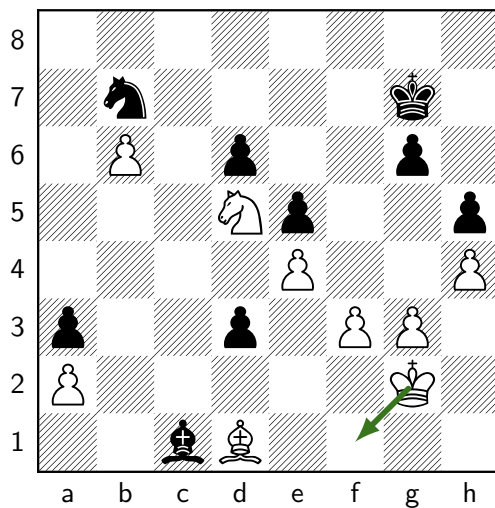


(d)

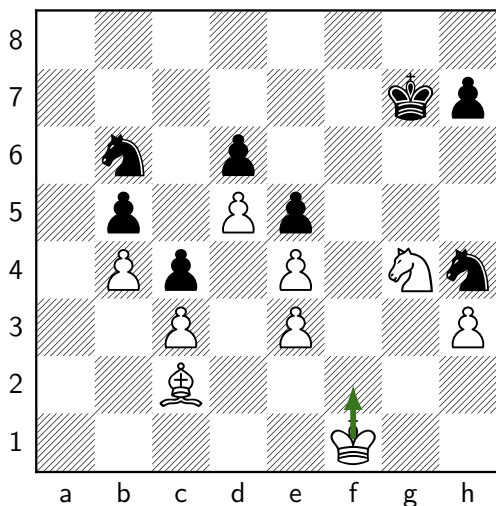
Figure 36: **Move 45, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1296` (ply 77) [structural distance $d=61$, QAP obj=6]: the master played Ke7; under μ our Ke1 plays the role of that game’s Kf1; cross-type roles: our Na3 as their Pg6, our Bb3 as their pg3; transferred move e1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#1514` (ply 85) [structural distance $d=58$, QAP obj=6]: the master played Kh5; under μ our Ke1 plays the role of that game’s Kg3; cross-type roles: our Pf4 as their Bd6, our Bb3 as their Pf5; transferred move e1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0024#2922` (ply 80) [structural distance $d=60$, QAP obj=4]: the master played Kf2; under μ our Ke1 plays the role of that game’s Kg1; cross-type roles: our Pf4 as their Nd2, our Bb3 as their na3; transferred move e1e2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Ke2 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



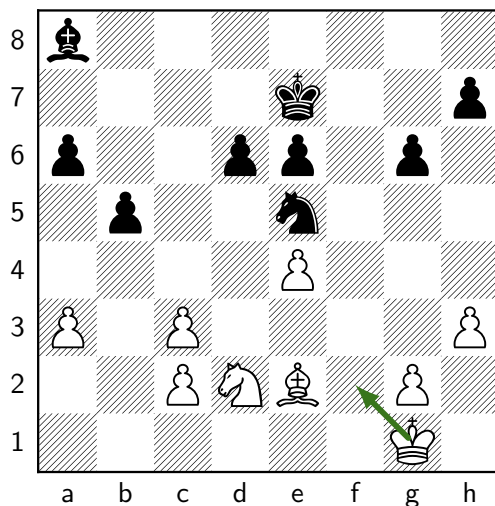
(a)



(b)

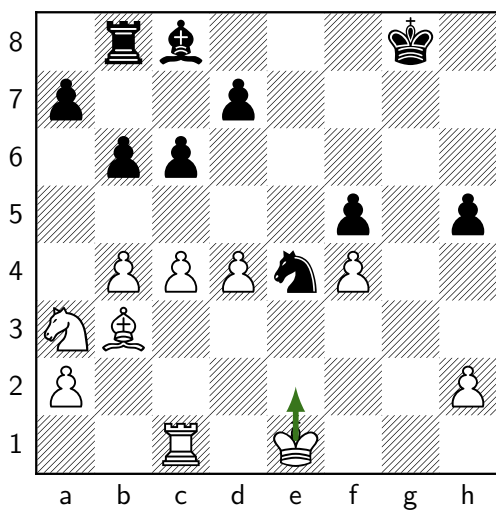


(c)

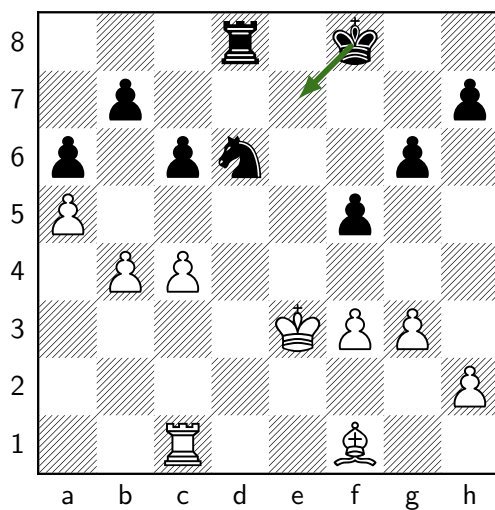


(d)

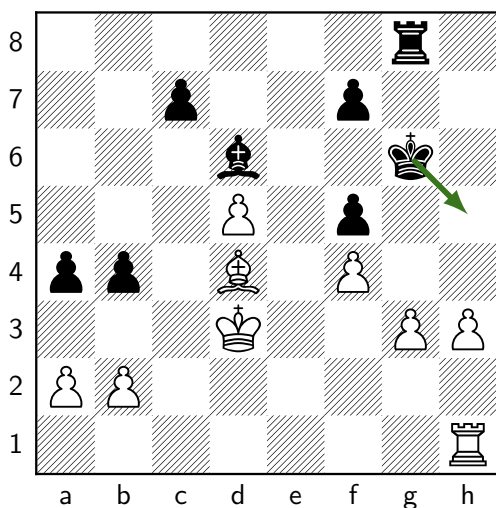
Figure 37: **Move 46, White: Ke1**. (a) Our position; Melanie plays Ke1 (e2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2354` (ply 86) [structural distance $d=62$, QAP obj=6]: the master played Kf1; under μ our Ke2 plays the role of that game’s Kg2; cross-type roles: our ne4 as their bc1, our Na3 as their Bd1; transferred move e2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#3081` (ply 78) [structural distance $d=58$, QAP obj=6]: the master played Kf2; under μ our Ke2 plays the role of that game’s Kf1; cross-type roles: our Pc4 as their Bc2, our bc8 as their Pb4; transferred move e2e1 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#2661` (ply 74) [structural distance $d=62$, QAP obj=5]: the master played Kf2; under μ our Ke2 plays the role of that game’s Kg1; cross-type roles: our Pd4 as their Nd2, our Pb4 as their Be2; transferred move e2e1 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Ke1 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



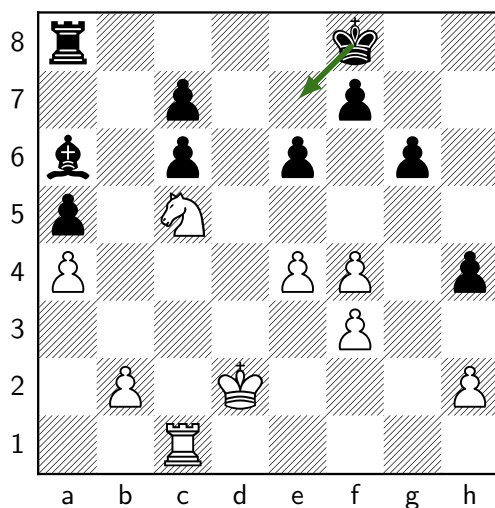
(a)



(b)



(c)



(d)

Figure 38: **Move 47, White: Ke2.** (a) Our position; Melanie plays Ke2 (e1e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#1296` (ply 77) [structural distance $d=61$, QAP obj=6]: the master played Ke7; under μ our Ke1 plays the role of that game’s Kf1; cross-type roles: our Na3 as their Pg6, our Bb3 as their pg3; transferred move e1e2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#1514` (ply 85) [structural distance $d=58$, QAP obj=6]: the master played Kh5; under μ our Ke1 plays the role of that game’s Kg3; cross-type roles: our Pf4 as their Bd6, our Bb3 as their Pf5; transferred move e1e2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0020#337` (ply 79) [structural distance $d=66$, QAP obj=5]: the master played Ke7; under μ our Ke1 plays the role of that game’s Kf1; cross-type roles: our ne4 as their Ba6, our Bb3 as their Pc6; transferred move e1e2 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Ke2 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.